An Ethical Riemann Hypothesis: A Formal Model for Quantifying AI Moral Fallibility under Complexity

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Abstract

As artificial intelligence systems increasingly participate in moral decision-making, we urgently need rigorous frameworks to understand when and how these systems fail structurally. This paper proposes the *Ethical Riemann Hypothesis* (ERH), a mathematical framework that quantifies error growth patterns in moral judgment systems through an analogy with prime number distribution in analytic number theory.

We define "ethical primes" as fundamental misjudgments that cannot be reduced to simpler cases, and introduce $\Pi(x)$, the counting function for ethical primes up to complexity level 100. By establishing a baseline expectation $B(x)$ analogous to the logarithmic integral, we characterize systematic deviations through the error term $E(x) = \Pi(x) - B(x)$. The ERH states that $|E(x)| \leq C \cdot x^{1/2+\varepsilon}$ for constants $C, \varepsilon > 0$, indicating that errors in "healthy" judgment systems grow at most sublinearly.

Through computational simulations comparing four judgment strategies—biased, noisy, conservative, and radical—on 2000 actions, we find that different systems exhibit markedly different error growth patterns: radical and biased judges show rapidly decreasing normalized error terms ($\alpha < -0.4$), while noisy and conservative judges display relatively flat patterns ($\alpha \approx -0.1$). Notably, remaining below the ERH-style worst-case bound is compatible with poor practical performance; the conservative judge's high mistake rate (61.1%) shows that ERH-style stability is, at most, a necessary structural condition rather than a guarantee of fairness or adequacy.

This framework provides a quantitative diagnostic for structural error accumulation in AI judgment systems. In practice, ERH can help detect long-tail instability and compare mitigation strategies, but it does not by itself establish fairness, calibration quality, or normative acceptability.

Keywords: Ethical Riemann Hypothesis, AI Ethics, Moral Judgment, Error Analysis, AI Fairness, Prime Number Theory, Structural Bias

1 Introduction

1.1 Motivation

The challenge of moral judgment is central to both human society and artificial intelligence systems. As AI increasingly participates in decisions affecting human welfare—from criminal justice and medical diagnosis to resource allocation and content moderation—we need rigorous frameworks for understanding when and how these systems fail [Jobin et al., 2019, Floridi et al., 2018].

Traditional approaches to AI ethics often focus on:

- **Rule-based frameworks**: Defining explicit moral rules (e.g., Asimov's Laws of Robotics)
- **Consequentialism**: Optimizing outcomes (e.g., utilitarian calculus)
- **Fairness metrics**: Measuring statistical parity, equal opportunity, and calibration [Hardt et al., 2016, Dwork et al., 2012]

While valuable, these approaches typically analyze individual cases or aggregate statistics. They provide limited insight into the *structural patterns* of misjudgment: How do errors accumulate as problems become more complex? Are there fundamental misjudgments that cascade through a system? When does a judgment system become systematically unreliable?

1.2 The Analogy with Prime Numbers

We propose an unexpected analogy: *the distribution of critical moral misjudgments resembles the distribution of prime numbers* [Riemann, 1859, Edwards, 1974].

In number theory:

- Prime numbers are the "atoms" of arithmetic—irreducible building blocks
- Their distribution appears irregular but follows deep patterns
- The Riemann Hypothesis characterizes the error in counting primes
- This error term encodes profound information about number-theoretic structure

In moral judgment:

- Some misjudgments are "ethical primes"—fundamental errors that don't reduce to simpler cases
- Their distribution across complexity levels appears irregular but may follow patterns
- An "Ethical Riemann Hypothesis" could characterize the error in counting these primes
- This error term encodes information about the judgment system's structural health

1.3 Contributions

This paper makes the following contributions:

1. **Mathematical Formalization**: We define action spaces, true moral values, judgment functions, and ethical primes in a rigorous mathematical framework (Section 3).

2. **The Ethical Riemann Hypothesis**: We state ERH precisely and interpret its meaning for judgment system quality (Section 4).

3. **Computational Framework**: We develop simulation tools to test ERH empirically across different judgment systems (Section 5).

4. **Empirical Results**: We demonstrate that biased, noisy, conservative, and radical judges exhibit distinct structural error patterns and substantially different practical risk profiles, even when all remain below the ERH-style worst-case bound in our main simulations (Section 6).

5. **Philosophical Implications**: We connect ERH to classical moral philosophy and discuss its implications for AI ethics (Section 8).

6. **AI Ethics Applications**: We discuss how ERH can be used as a structural guardrail within broader audit and design workflows for AI systems (Section 9).

1.4 Target Audience and Use Cases

This paper is written for researchers and practitioners across multiple domains: **AI ethics** researchers seeking quantitative frameworks for evaluating moral judgment systems, **machine learning fairness** practitioners who need to complement traditional fairness metrics with structural health indicators, **philosophy of technology** scholars interested in formal models of ethical reasoning, and **AI governance and policy** stakeholders who require operational criteria for regulatory oversight. Readers can use the ERH framework to: (1) *audit* existing AI systems for structural degradation patterns, (2) *treat* ERH-style bounds as guardrails when designing or revising systems, and (3) *monitor* deployment for signs of worsening long-tail instability. The framework provides a shared quantitative language for one aspect of system behavior—complexity-linked error accumulation—that can be combined with fairness, calibration, and domain-specific evaluation.

1.5 Paper Organization

The remainder of this paper is organized as follows. Section 2 reviews related work and theoretical foundations. Section 3 develops the mathematical formalization. Section 4 states the Ethical Riemann Hypothesis and its variants. Section 5 describes our computational framework. Section 6 presents experimental results. Section 8 discusses philosophical implications. Section 9 discusses applications to AI ethics. Section 10 concludes and outlines future work.

2 Related Work and Theoretical Foundations

2.1 AI Judgment Errors and Ethical Risk

The literature on AI ethics has extensively documented various forms of bias and unfairness in algorithmic systems [Mehrabi et al., 2021, Barocas and Selbst, 2016, O'Neil, 2016]. However, most approaches focus on statistical measures of fairness (e.g., demographic parity, equalized odds) or individual case analysis, rather than structural patterns of error accumulation.

Binns [2018] argues that fairness in machine learning should draw from political philosophy, particularly theories of distributive justice. Selbst et al. [2019] emphasizes that fairness must be understood within sociotechnical systems, not as isolated algorithmic properties. While these works provide important theoretical foundations, they lack quantitative frameworks for measuring structural degradation.

2.2 Mathematical Models in AI Ethics

Several researchers have applied mathematical formalisms to ethical problems. Dwork et al. [2012] introduced formal definitions of fairness through awareness, using metric spaces to quantify similarity. However, these approaches focus on static properties rather than dynamic error growth patterns. In formal semantics and logics of agency, Cohen and Levesque proposed a classic model that treats intention as choice with commitment [Cohen and Levesque, 1990], laying groundwork for subsequent work on deontic and intentional logics.

2.3 Metaphorical Models in Philosophy

The use of mathematical metaphors in philosophy has a rich history. Control theory metaphors have been applied to ethics [Floridi et al., 2018], and game theory has been used to model moral dilemmas. Our approach extends this tradition by using analytic number theory as a lens for understanding moral judgment systems.

2.4 The Riemann Hypothesis in Number Theory

The classical Riemann Hypothesis, proposed by Riemann [1859], concerns the distribution of zeros of the Riemann zeta function $\zeta(s)$. Montgomery [1973] showed that the pair correlation of zeros follows specific patterns, suggesting deep structure in prime distribution. Granville and Soundararajan [2007] further developed connections between character sums and prime distribution.

While the Riemann Hypothesis remains unproven, its implications for prime number distribution are profound. We adapt this framework to ethical primes, recognizing that the analogy is heuristic but potentially illuminating.

2.5 Gap Analysis

Current approaches to AI ethics suffer from several limitations:

- **Lack of structural analysis**: Most methods analyze individual cases or aggregate statistics, missing patterns of error accumulation

- **No complexity-aware metrics**: Existing fairness metrics don't account for how errors scale with problem complexity
- **Insufficient formalization**: Few frameworks provide rigorous mathematical models for ethical error patterns
- **Missing structural warning criteria**: There are few quantitative criteria for when a judgment system becomes systematically unreliable as case complexity increases

Our ERH framework addresses these gaps by providing a structural, complexity-aware, formally rigorous, and diagnostic model of ethical error patterns.

Table 1: Comparison of Existing Methods with ERH Framework

Feature	**Existing Methods**	**ERH Framework**
Analysis Level	Individual/Aggregate	Structural
Complexity Awareness	No	Yes
Mathematical Rigor	Limited	High
Structural Warning Signal	Limited	Yes
Error Growth Model	None	Sublinear bound

2.6 Comparison to Reliability, Calibration, and Fairness Metrics

Our ERH framework is closely related to, but conceptually distinct from, existing reliability and fairness metrics in machine learning. Classical *reliability* measures such as the Brier score and expected calibration error (ECE) evaluate how well predicted probabilities align with empirical frequencies on a fixed prediction task. They are fundamentally *pointwise or distributional* diagnostics: given a pool of predictions and labels, they assess whether a model is well-calibrated overall, without explicitly tracking how errors accumulate as the problem becomes more complex.

By contrast, ERH is explicitly *complexity-aware*. Instead of asking whether predictions are calibrated at a single level of difficulty, we ask how the count of fundamental misjudgments—ethical primes—grows as we move along a complexity axis. The error term $E(x) = \Pi(x) - B(x)$ and its conjectured bound $|E(x)| \leq Cx^{1/2+\varepsilon}$ provide a global constraint on how quickly structural failure can accumulate across complexity buckets. This makes ERH particularly sensitive to long-tail, high-complexity regimes where standard calibration scores may remain acceptable while rare but catastrophic errors proliferate.

Similarly, many *fairness* metrics (e.g., demographic parity, equal opportunity, calibration within groups) are defined on protected subpopulations but remain largely agnostic to how error patterns evolve with task complexity. They tell us whether a system is fair *on average* for each group, but not whether it exhibits structural degradation concentrated in high-stakes, high-complexity regions of the action space. ERH complements these metrics by providing:

- a **structural view** of misjudgment that tracks how ethical primes accumulate as complexity increases;

- a **growth-exponent** perspective that distinguishes systems with benign error scaling from those whose error structures explode in the tail;
- a **macro-level diagnostic** for comparing judgment systems or mitigation strategies, even when average accuracy and fairness scores look similar.

In short, existing reliability and fairness measures characterize how a system behaves at a fixed level of difficulty or across demographic groups, whereas ERH characterizes how its *structural error landscape* evolves along the complexity axis. This distinction underlies the role of ERH as a complementary auditing tool rather than a replacement for standard metrics.

3 Model Construction

3.1 Action Space

Definition 3.1 (Action Space)**.** An *action space* is a finite set $\mathcal{A} = \{a_1, a_2, \ldots, a_N\}$ where each action a_i has the following attributes:

- **Complexity** $c(a) \in \mathbb{N}$: A positive integer representing the decision's complexity (information requirements, number of stakeholders, uncertainty, etc.)
- **True Moral Value** $V(a) \in \mathbb{R}$: The "ground truth" ethical evaluation, typically normalized to $[-1, 1]$ where -1 is maximally harmful, 0 is neutral, and $+1$ is maximally beneficial
- **Importance Weight** $w(a) \in \mathbb{R}^+$: The significance of the action (number of people affected, magnitude of consequences, etc.)

Remark 3.2. The true moral value $V(a)$ represents an idealized "god's-eye view" or consensus among perfect moral reasoners. In practice, this is inaccessible and must be approximated. In our simulations, we define $V(a)$ explicitly as ground truth.

3.2 Judgment Systems

Definition 3.3 (Judgment System)**.** A *judgment system* $\mathcal{J}$ is a function that assigns a moral evaluation to each action:

$$\mathcal{J} : \mathcal{A} \to \mathbb{R}$$

We denote $J(a) = \mathcal{J}(a)$ as the judgment of action a.

Definition 3.4 (Judgment Error)**.** The *error* of judgment $\mathcal{J}$ on action a is:

$$\Delta(a) = J(a) - V(a)$$

A judgment is considered a *mistake* if $|\Delta(a)| > \tau$ for some threshold $\tau > 0$.

We define a binary mistake indicator:

$$M(a) = \begin{cases} 1 & \text{if } |\Delta(a)| > \tau \\ 0 & \text{otherwise} \end{cases}$$

3.3 Ethical Primes

Definition 3.5 (Ethical Prime)**.** An action $a \in \mathcal{A}$ is an *ethical prime* if:

1. $M(a) = 1$ (it is misjudged)
2. $w(a)$ is large (high importance, typically in top quantile)
3. a is "structurally fundamental": correcting this error would significantly reduce overall misjudgment

Let $\mathcal{P} \subseteq \mathcal{A}$ denote the set of ethical primes. In practice, we select $\mathcal{P}$ by filtering misjudged actions by importance quantile and complexity range.

Proposition 3.6 (Stability of Ethical Prime Set)**.** *Given an action space $\mathcal{A}$ and a judgment system $\mathcal{J}$, let the importance threshold be w_0. Then the ethical prime set $\mathcal{P}$ satisfies:*

$$|\mathcal{P}| \leq \min\left(|\{a \in \mathcal{A} : M(a) = 1\}|, |\{a \in \mathcal{A} : w(a) \geq w_0\}|\right)$$

Furthermore, if the judgment system satisfies a uniform error bound $|\Delta(a)| \leq \delta$ for all $a \in \mathcal{A}$ with error threshold $\tau > \delta$, then $\mathcal{P} = \emptyset$. Conversely, if there exists some complexity level c_0 such that for all actions with $c(a) \geq c_0$, the probability that $|\Delta(a)| > \tau$ and $w(a) \geq w_0$ is $p > 0$, then the expected number of ethical primes is at least $p \cdot |\{a \in \mathcal{A} : c(a) \geq c_0, w(a) \geq w_0\}|$.

Proof Sketch. The first inequality follows directly from the definition of ethical primes: they must be both mistakes ($M(a) = 1$) and high importance ($w(a) \geq w_0$). If the uniform error bound holds, then $|\Delta(a)| \leq \delta < \tau$ for all actions, so $M(a) = 0$ for all a, resulting in $\mathcal{P} = \emptyset$. The final expectation follows from basic probability theory: if each qualifying action becomes an ethical prime with probability p, the expected count is the product of probability and quantity. □

3.4 Theoretical Distinction of Judge Types

Different types of judgment systems exhibit distinct theoretical characteristics in their error structures. We can establish error bounds and expected behaviors for each type:

Proposition 3.7 (Error Structure of Biased Judge)**.** *For a biased judge with monotone increasing bias function $b(c)$ and $b(0) = 0$, noise variance σ^2, the expected error satisfies:*

$$\mathbb{E}[|\Delta(a)|] \geq |b(c(a))| - \sigma\sqrt{\frac{2}{\pi}}$$

If $b(c) = b_0 \cdot c^\beta$ for some $b_0 > 0, \beta > 0$, then error grows with complexity, and ethical primes are more likely to occur in higher $c(a)$ ranges.

Proposition 3.8 (Error Structure of Noisy Judge)**.** *For a noisy judge with noise variance $\sigma^2(c)$ that may vary with complexity, the variance of error is:*

$$Var[\Delta(a)] = \sigma^2(c(a))$$

If $\sigma(c) = \sigma_0 \cdot c^\gamma$ for $\sigma_0 > 0, \gamma \geq 0$, then high-complexity cases have larger error variability. When $\gamma > 0$, the error growth exponent α tends toward 0, as error is primarily dominated by randomness rather than systematic patterns.

Proposition 3.9 (Error Structure of Conservative Judge). *For a conservative judge with shrinkage factor $\lambda(c) \in [0, 1]$ that increases monotonically with complexity, the expected error is approximately:*

$$\mathbb{E}[|\Delta(a)|] \approx |V(a)| \cdot \lambda(c(a))$$

The conservative strategy causes all judgments to shrink toward neutral, so low-complexity cases ($|V(a)|$ near 1) may exhibit large errors, while high-complexity ambiguous cases ($|V(a)| \approx 0$) have small errors. This explains why conservative judges have the highest mistake rate but near-zero error growth exponent.

Proposition 3.10 (Error Structure of Radical Judge). *For a radical judge with amplification factor $\alpha > 1$, the error is:*

$$\Delta(a) = (\alpha - 1) \cdot V(a) + \mathcal{N}(0, \sigma^2)$$

When $|V(a)|$ is large, amplification causes significant errors; when $|V(a)| \approx 0$, error is primarily dominated by noise. Thus radical judges perform poorly on low-complexity clear cases but may provide clearer discriminative signals in high-complexity ambiguous cases, leading to decreasing error as complexity increases.

3.5 Counting Functions

Definition 3.11 (Ethical Prime Counting Function). Define $\Pi(x)$ as the number of ethical primes with complexity at most x:

$$\Pi(x) = \#\{p \in \mathcal{P} : c(p) \leq x\}$$

Definition 3.12 (Baseline Function). The *baseline function* $B(x)$ represents the expected number of ethical primes up to complexity x under some smooth model. We consider several forms:

1. **Linear**: $B(x) = \alpha x$ for some $\alpha > 0$
2. **Prime Theorem Analog**: $B(x) = \beta \frac{x}{\log x}$ for some $\beta > 0$, directly analogous to the Prime Number Theorem
3. **Logarithmic Integral**: $B(x) = \beta \cdot \mathrm{Li}(x)$ where $\mathrm{Li}(x) = \int_2^x \frac{dt}{\log t}$
4. **Power Law**: $B(x) = \gamma x^\delta$ for parameters $\gamma, \delta > 0$

Definition 3.13 (Error Term). The *error term* is:

$$E(x) = \Pi(x) - B(x)$$

This measures how the actual distribution of ethical primes deviates from the baseline expectation.

3.6 Example: Computing $\Pi(x)$

Consider a simple example with three actions:

- a_1: $c(a_1) = 5$, $V(a_1) = 0.8$, $w(a_1) = 10$, $|\Delta(a_1)| = 0.4$ (mistake)

- a_2: $c(a_2) = 15$, $V(a_2) = -0.6$, $w(a_2) = 8$, $|\Delta(a_2)| = 0.5$ (mistake)
- a_3: $c(a_3) = 20$, $V(a_3) = 0.3$, $w(a_3) = 12$, $|\Delta(a_3)| = 0.2$ (not a mistake)

If $\tau = 0.3$ and we select ethical primes as top 50% by importance among mistakes, then $\mathcal{P} = \{a_1, a_2\}$ (both are mistakes, and a_1 has highest importance). Then:

$$\begin{aligned}
\Pi(10) &= 1 \quad \text{(only } a_1 \text{ has } c \leq 10\text{)} \\
\Pi(15) &= 2 \quad \text{(both } a_1 \text{ and } a_2 \text{ have } c \leq 15\text{)} \\
\Pi(20) &= 2 \quad \text{(both have } c \leq 20\text{)}
\end{aligned}$$

4 The Ethical Riemann Hypothesis

4.1 Statement of ERH

Theorem 4.1 (Ethical Riemann Hypothesis (ERH))**.** *Let $\mathcal{J}$ be a judgment system and let $E(x) = \Pi(x) - B(x)$ be the error term for its ethical prime distribution. The judgment system satisfies the* Ethical Riemann Hypothesis *if there exist constants $C, \varepsilon > 0$ such that:*

$$|E(x)| \leq C \cdot x^{1/2+\varepsilon}$$

for all x in the complexity range. This bound $|E(x)| \leq C \cdot x^{1/2+\varepsilon}$ is called the ERH-style bound*.*

Remark 4.2 (ERH as an Empirical Diagnostic Hypothesis)*.* In contrast to the classical Riemann Hypothesis, we do *not* claim to prove or refute ERH as a formal theorem about an underlying mathematical object. Throughout this paper, ERH is treated as an *empirical diagnostic hypothesis*: an ideal error-growth bound inspired by analytic number theory, against which we test concrete (simulated or real-world) judgment systems. Our contribution is to propose this ERH-style bound as a diagnostic lens and to develop computational tools that estimate error growth and check whether observed systems are consistent with the bound within reasonable finite-sample slack.

Remark 4.3 (ERH as a Necessary but Not Sufficient Condition)*.* The ERH-style bound $|E(x)| \leq C \cdot x^{1/2+\varepsilon}$ provides a *necessary condition* for structural health of a judgment system: it ensures that error growth remains sublinear and bounded. However, satisfying this bound alone is *not sufficient* to guarantee that a system is morally acceptable. A judgment system must also maintain low overall error rates and avoid systematic biases. As demonstrated in our experimental results (Section 6), a system can satisfy the ERH-style bound while still exhibiting unacceptably high mistake rates (e.g., the Conservative Judge with 61.1% mistake rate). Therefore, the ERH-style bound should be evaluated alongside other metrics such as overall accuracy, fairness measures, and domain-specific quality criteria.

4.2 Intuitive Interpretation

The ERH states that the cumulative error in predicting critical misjudgments grows at most like $\sqrt{x}$ (up to a small factor x^{ε}). This is a *sublinear* growth rate, meaning:

- As problem complexity increases, the judgment system does not lose control

- Errors accumulate slowly enough to remain structurally auditable
- The system maintains structural integrity across scales

4.3 Probabilistic Setting

To connect ERH with a probabilistic view of judgment systems, we model ethical prime generation as a random process over complexity levels. For each integer complexity level $c \in \{1, 2, \dots\}$, let $N(c)$ denote the (possibly random) number of actions with complexity exactly c, and let

$$X_c = \#\{a \in \mathcal{A} : c(a) = c, \ a \text{ is an ethical prime}\}$$

be the random number of ethical primes at complexity c. Then the cumulative prime count and baseline can be written as

$$\Pi(x) = \sum_{c \le x} X_c, \qquad B(x) = \sum_{c \le x} \mu_c,$$

where $\mu_c = \mathbb{E}[X_c]$ is the expected number of ethical primes at level c under some smooth generative model. The error term becomes

$$E(x) = \sum_{c \le x} (X_c - \mu_c).$$

We make the following assumptions:

(A1) (*Controlled growth of candidate actions*) There exist constants $C_0, d > 0$ such that almost surely

$$0 \le X_c \le N(c) \le C_0 c^d \quad \text{for all } c \ge 1.$$

(A2) (*Decaying prime rate*) There exist constants $C_1, \beta > 0$ such that

$$\mu_c = \mathbb{E}[X_c] \le C_1 c^{-\beta} N(c) \quad \text{for all sufficiently large } c.$$

Intuitively, the probability that a high-complexity action becomes an ethical prime decays as a power of c.

(A3) (*Approximate independence*) Conditional on $N(c)$, the variables X_c are independent across complexities, and each X_c can be viewed as a sum of at most $N(c)$ bounded contributions (e.g., Bernoulli indicators for each action).

Assumptions (A1)–(A3) capture the idea that while there may be many high-complexity actions, ethical primes become increasingly rare, and prime occurrences at different complexity levels do not exhibit pathological long-range dependence.

4.4 A Non-trivial Probabilistic Theorem

Under the probabilistic setting above, we can show that ERH-type bounds arise naturally when ethical primes become sufficiently rare at high complexity.

Theorem 4.4 (Probabilistic ERH Bound under Decaying Prime Rate)**.** *Suppose (A1)–(A3) hold with parameters* $C_0, d, C_1, \beta > 0$. *Assume moreover that* $\beta > 1$ *and that for some* $\varepsilon \in (0, \frac{\beta-1}{2})$ *the following variance bound holds:*

$$\mathrm{Var}(X_c) \leq C_2 c^{d-\beta} \quad \textit{for all sufficiently large } c$$

for a constant $C_2 > 0$. *Then there exists a constant* $C > 0$ *(depending on* $C_0, C_1, C_2, d, \beta, \varepsilon$*) such that with probability at least* $1 - O(x^{-\varepsilon})$ *we have*

$$|E(x)| \leq C\, x^{1/2+\varepsilon} \quad \textit{for all sufficiently large } x.$$

In particular, under these assumptions the judgment system satisfies an ERH-type sublinear growth bound for the error term with high probability.

Proof Sketch. Write $Y_c = X_c - \mu_c$, so that $E(x) = \sum_{c \leq x} Y_c$ and $\mathbb{E}[Y_c] = 0$. By (A1) and (A2), $|Y_c| \leq X_c + \mu_c \leq 2C_0 c^d$ for all large c, and by the variance assumption we have

$$\sum_{c \leq x} \mathrm{Var}(Y_c) = \sum_{c \leq x} \mathrm{Var}(X_c) \leq C_2 \sum_{c \leq x} c^{d-\beta} = O(x^{d-\beta+1}).$$

Since $\beta > 1$, the exponent $d - \beta + 1$ is strictly less than d.

Using Bernstein's inequality (or another standard concentration inequality for sums of independent, bounded random variables), for any $t > 0$ we obtain

$$\mathbb{P}\big(|E(x)| \geq t\big) \leq 2 \exp\left(- \frac{t^2/2}{\sum_{c \leq x} \mathrm{Var}(Y_c) + Mt} \right),$$

where M is a bound on $|Y_c|$. Substituting the bounds above and choosing $t = Kx^{1/2+\varepsilon}$ for a suitable constant $K > 0$, the exponent becomes polynomially small in x provided $\varepsilon < (\beta - 1)/2$. Thus we obtain

$$\mathbb{P}\big(|E(x)| \geq Kx^{1/2+\varepsilon}\big) = O(x^{-\varepsilon})$$

for large x. Applying a union bound over dyadic scales of x yields the claimed high-probability bound for all sufficiently large x simultaneously. □

This theorem is intentionally modest in scope: it does not claim that all reasonable judgment systems must satisfy ERH, but rather that *under explicit probabilistic assumptions about how ethical primes thin out at high complexity*, an ERH-type bound on $E(x)$ follows from standard concentration arguments. In this sense, ERH can be understood as a mathematically natural regularity condition for judgment systems whose structural failures do not explode in the tail.

4.5 Comparison with Classical Riemann Hypothesis

4.6 Ethical Zeta Function

We can define an "ethical zeta function" via generating functions:

Definition 4.5 (Ethical Zeta Function)**.** Let $m(n)$ be the number (or weighted count) of ethical primes at complexity level n. Define:

$$\zeta_E(s) = \sum_{n=1}^{N} \frac{m(n)}{n^s}$$

for $s \in \mathbb{C}$ with $\mathrm{Re}(s) > 1$.

Table 2: Analogy between Classical RH and ERH

Number Theory	**Ethical Judgment**
Natural number n	Action complexity c
Prime p	Ethical prime (critical misjudgment)
$\pi(x)$ (prime counting)	$\Pi(x)$ (ethical prime counting)
$\mathrm{Li}(x) \sim \frac{x}{\log x}$	$B(x) \sim \frac{x}{\log x}$
$E(x) = \pi(x) - \mathrm{Li}(x)$	$E(x) = \Pi(x) - B(x)$
RH: $\lvert E(x)\rvert = O(x^{1/2} \log x)$	ERH: $\lvert E(x)\rvert = O(x^{1/2+\varepsilon})$
$\zeta(s)$ Riemann zeta function	$\zeta_E(s)$ Ethical zeta function
Zeros of $\zeta(s)$	Zeros of $\zeta_E(s)$

4.7 Analytic Properties of the Ethical Zeta Function

Although the ethical zeta function $\zeta_E(s)$ is a finite sum (unlike the infinite series of the classical Riemann zeta function), we can still analyze its basic analytic properties and establish heuristic connections with the error term $E(x)$.

Proposition 4.6 (Domain of Convergence and Analyticity)**.** *The ethical zeta function $\zeta_E(s)$ is defined for all $s \in \mathbb{C}$ (since it is a finite sum), but it exhibits structural characteristics similar to the classical zeta function in the region $Re(s) > 1$. If $m(n) = O(n^\delta)$ for some $\delta < 1$, then in the region $Re(s) > \delta + 1$, the growth of $\zeta_E(s)$ is primarily dominated by low-complexity terms.*

Proposition 4.7 (Heuristic Connection between Zero Distribution and Error Term)**.** *(Heuristic, non-rigorous) If the zeros of the ethical zeta function $\zeta_E(s)$ cluster near some critical line (e.g., $Re(s) = \sigma_0$), then the growth rate of the error term $E(x)$ may relate to σ_0:*

$$|E(x)| \sim x^{\sigma_0} \quad \textit{(heuristic)}$$

Specifically, if zeros cluster near $Re(s) = 1/2$, this may indicate that $|E(x)| = O(x^{1/2+\varepsilon})$, consistent with the ERH-style worst-case bound. This is a heuristic analogy rather than a rigorous theorem, as the connection between zeros of the classical zeta function and the error term in prime distribution involves sophisticated analytic number theory techniques (such as Perron's formula and Mellin transforms), which our finite-sum version does not possess to the same degree of theoretical structure.

Remark 4.8 (Heuristic Value and Limitations of the Analogy). The analogy between the ethical zeta function and the classical Riemann zeta function has heuristic value but important differences:

- **Finitude**: $\zeta_E(s)$ is a finite sum, so there is no analytic continuation, functional equation, or critical strip concept, which are key features of the classical zeta function.
- **Structural differences**: The distribution of ethical primes may not possess the deep number-theoretic structure of prime distribution (such as the prime number theorem in arithmetic progressions).
- **Heuristic value**: Nevertheless, zero analysis may still reveal periodic patterns in error distribution, and spectral analysis (corresponding to discrete Fourier transform) can identify correlations between complexity levels.

The distribution of zeros of $\zeta_E(s)$ in the complex plane encodes information about the "periodicity" or "regularity" of ethical prime distribution. If zeros cluster near a critical line (analogous to $\mathrm{Re}(s) = 1/2$ for the Riemann zeta), this suggests deep structural regularity in the judgment errors. However, we must recognize this as a heuristic analogy rather than a rigorous number-theoretic correspondence.

5 Simulation Design and Experimental Methods

5.1 Action Space Generation

We generate actions with complexity sampled from:

- **Uniform distribution**: $c(a) \sim \mathrm{Uniform}(1, C_{\max})$
- **Zipf distribution**: $c(a) \sim \mathrm{Zipf}(\alpha)$ (more realistic, many simple cases, few complex)
- **Power law**: $c(a) \sim x^{-\gamma}$

True moral values $V(a)$ are generated with *complexity-dependent ambiguity*:

- Low complexity: clear values (near ± 1)
- High complexity: ambiguous values (near 0)

This reflects the intuition that simple moral cases are often clear-cut, while complex cases involve multiple competing considerations.

5.2 Judgment System Models

We implement five archetypal judges:

1. **BiasedJudge**: Systematic bias $b(c)$ increasing with complexity:
$$J(a) = V(a) + b_0 \cdot f(c(a)) + \mathcal{N}(0, \sigma^2)$$
where $f(c)$ is monotone increasing.

2. **NoisyJudge**: High random noise:
$$J(a) = V(a) + \mathcal{N}(0, \sigma^2(c))$$
where noise scales with complexity.

3. **ConservativeJudge**: Shrinks toward neutral:
$$J(a) = (1 - \lambda(c))V(a) + \mathcal{N}(0, \sigma^2)$$
where $\lambda(c) \in [0, 1]$ increases with complexity.

4. **RadicalJudge**: Amplifies extremes:
$$J(a) = \alpha \cdot V(a) + \mathcal{N}(0, \sigma^2)$$
where $\alpha > 1$.

5. **QuantumJudge**: Superposition-based judgment via quantum measurement. Let $\theta = c(a)/100 \cdot \pi$; apply $\mathrm{Rx}(\theta)$ rotation to $|0\rangle$. Measurement yields $J(a) = 2|\alpha|^2 - 1 \in [-1, 1]$, where $|\alpha|^2$ is the probability of measuring "ethical" ($|0\rangle$). Uses qiskit-aer when available; otherwise NumPy fallback for deterministic reproducibility.

5.3 Analysis Pipeline

For each judgment system:

1. Generate action space with $N = 2000$ actions
2. Evaluate all actions with judge $\mathcal{J}$
3. Compute errors $\Delta(a)$ and identify mistakes
4. Select ethical primes $\mathcal{P}$ (top 10% by importance)
5. Compute $\Pi(x)$, $B(x)$, $E(x)$ for $x \in [1, 100]$
6. Fit power law: $|E(x)| \sim Cx^{\alpha}$
7. Test if $\alpha \approx 0.5$ (ERH satisfied)

5.4 Parameter Settings

Table 3: Simulation Parameters

Parameter	**Value**
Number of actions (N)	2000
Complexity range	[1, 100]
Complexity distribution	Zipf($\alpha = 2.0$)
Error threshold (τ)	0.3
Importance quantile for primes	0.9
$X_{\max}$ for analysis	100
Random seed	42

5.5 Reproducibility and Statistical Stability

To ensure experimental reproducibility, we adopted the following measures:

- **Fixed random seed**: All simulations use a fixed random seed ($seed = 42$), ensuring consistency in action space generation, noise generation, and judgment evaluation.
- **Parameter documentation**: All judge types and their parameter settings (as shown in Table 3) are explicitly documented for reproducibility and verification.
- **Action generation process**: Action space generation follows these steps:
    1. Complexity $c(a)$ is sampled from a Zipf distribution: $c(a) \sim \text{Zipf}(\alpha = 2.0)$ with range $[1, 100]$
    2. True moral values $V(a)$ are generated with complexity-dependent ambiguity: $V(a) = \text{sign}(\mathcal{N}(0, 1)) \cdot \min(1, |\mathcal{N}(0, 1)| \cdot (1 - c(a)/100))$, where low-complexity cases ($c(a) \approx 1$) tend toward ± 1, and high-complexity cases ($c(a) \approx 100$) tend toward 0

3. Importance weights $w(a)$ are sampled from a log-normal distribution, $\log w(a) \sim \mathcal{N}(\mu = 2, \sigma = 1)$, so a small number of actions carry high importance

- **Judge system implementation details**:
  - **Biased Judge**: $J(a) = V(a) + 0.2 \cdot (c(a)/100)^2 + \mathcal{N}(0, 0.1^2)$, bias grows quadratically with complexity
  - **Noisy Judge**: $J(a) = V(a) + \mathcal{N}(0, (0.3 \cdot \sqrt{c(a)/100})^2)$, noise standard deviation grows with square root of complexity
  - **Conservative Judge**: $J(a) = (1 - 0.5 \cdot c(a)/100) \cdot V(a) + \mathcal{N}(0, 0.1^2)$, shrinkage factor increases linearly with complexity
  - **Radical Judge**: $J(a) = 1.5 \cdot V(a) + \mathcal{N}(0, 0.1^2)$, fixed amplification factor
  - **Quantum Judge**: $\mathrm{Rx}(\theta)$ with $\theta = c(a)/100 \cdot \pi$, shots=512; qiskit-aer or NumPy fallback

- **Error growth exponent fitting**: We fit $|E(x)| \sim Cx^\alpha$ using linear regression in log-log scale, where α is the error growth exponent. Fitting is performed over $x \in [10, 100]$ (excluding low-complexity region to reduce small-sample effects), using least squares estimation. We also report bootstrap confidence intervals for α obtained by resampling the $(x, |E(x)|)$ pairs, and we treat R^2 as a supplementary diagnostic rather than a primary model selection criterion.

- **Experimental repeatability**: Although the figures in the main text are based on a single representative run, we verified statistical stability across independent runs with 200 different random seeds. For each judge, we re-estimate the growth exponent α across all seeds and compute mean, standard deviation, coefficient of variation (CV), and empirical 95% confidence intervals. The CV of α remains below 0.15 for all judges and the confidence intervals are narrow, indicating good consistency. Table 4 summarizes the stability metrics across multiple random seeds.

Table 4: Statistical stability of error growth exponent α across multiple random seeds. All judges exhibit low coefficient of variation (CV < 0.15), indicating stable growth exponents across random seeds.

Judge Type	**mean(α)**	**std(α)**	**CV**
Biased	-0.629	0.042	0.067
Noisy	-0.171	0.018	0.105
Conservative	-0.046	0.005	0.109
Radical	-0.452	0.031	0.069

5.6 User Study of ERH Visualizations

To assess whether our ERH-style plots are interpretable for humans, we conducted a small visualization usability check with eight technically trained participants (graduate students and engineers). Participants were shown pairs of judges and parameter settings using our error-growth plots (including α and bound-violation indicators) and were asked to (i) rank which judge exhibits higher long-tail risk and (ii) identify which parameter setting appears most structurally fragile. Across all trials, participants' rankings agreed with the ordering induced by our ERH metrics in 87.5% of cases, and the median interpretability rating of the plots was 4 out of 5 on a Likert scale.

These results suggest that ERH-style visualizations are not only mathematically well-defined but also usable as risk-communication tools for practitioners. In particular, the combination of growth exponents, ERH-style bound satisfaction, and long-tail error profiles enables users to distinguish between judges that have similar average accuracy and fairness scores but very different structural error behaviors.

5.7 Ethical Considerations

Our simulations operate on synthetic action spaces and anonymized, tabular real-world case studies with no personal identifiers or sensitive free text. The framework evaluates *judgment rules* (e.g., abstract "judges") rather than individual people, and all fairness-related analyses are performed at an aggregate group level. Nevertheless, ERH-style diagnostics could be misused if treated as a single scalar "fairness score" for high-stakes deployment decisions. We therefore recommend using ERH metrics only as diagnostic tools, in conjunction with established fairness and accountability criteria, and not as certification mechanisms. All code and configuration files necessary to reproduce our experiments (including hyperparameters, random seeds, and dataset preprocessing steps) are provided in the public repository accompanying this paper.

All judges exhibit low coefficient of variation (CV < 0.15), indicating stable growth exponents across random seeds. This demonstrates that the observed error growth patterns are robust to random initialization and not artifacts of a particular seed choice.

5.8 Data and Code Availability

Complete code, simulation scripts, and generated figures for this study are publicly available to promote reproducibility and further research:

- **Code repository**: All simulation code, analysis scripts, and visualization tools are available at `https://github.com/dennislee928/Ethic-Latex`. For reviewers, the main supplementary entry points are `docs/EXPERIMENT_REPORTS.md` and `simulation/output/`.
- **Data outputs**: Numerical results from simulations (including judge metrics, error sequences, spectrum data, stability-of-α tables across random seeds, and quantum Hilbert space metrics) are saved under `simulation/output/`, including:
  - `results_summary.txt`: Statistical summary for all judges
  - `judge_comparison_report.md`: Detailed comparison report
  - `spectrum_data.json` and `zeros_data.json`: Spectrum and zero analysis data
  - `quantum_hilbert_results.json`: Consensus state, system coherence, Von Neumann entropy from quantum Hilbert space simulation
  - `*.csv`: Parameter sensitivity analysis results

- **Figure generation**: Core figures are generated by `simulation/generate_all_figures.py` and `scripts/run_quantum_hilbert_figures.py`. PDF figures are written to `simulation/output/figures/`, and the quantum Hilbert run also emits `latest_quantum_circuit.png` and `latest_quantum_distribution.png`.

- **Environment dependencies**: All necessary packages are listed in `requirements.txt`, with Python 3.10 or higher recommended. Main dependencies include `numpy`, `scipy`, `matplotlib`, `pandas`, and `jupyter`.

- **Verification and reproduction**: Readers can reproduce experiments by executing:

  ```
  cd simulation
  export PYTHONPATH=$PYTHONPATH:$(pwd)
  python generate_all_figures.py
  ```

  or use the provided Jupyter notebooks (`simulation/notebooks/`) for interactive exploration.

5.9 Real-World Data Acquisition

Real-data case studies (Adult Income, COMPAS) require datasets that are not bundled in the repository. Table 5 lists the data sources and output paths.

Table 5: Real-world data sources, URLs, and output paths

Dataset	**Source**	**Output Path**
Adult Income (train)	UCI ML Repository	`data/real_world/adult.data`
Adult Income (test)	UCI ML Repository	`data/real_world/adult.test`
Adult Income (processed)	—	`data/adult.csv` (after conversion)
COMPAS	ProPublica GitHub	`data/compas-scores-two-years.csv`

Fetch and conversion steps:

1. Run the repository's data-fetch script to download Adult from the UCI ML Repository and COMPAS from ProPublica into `data/`.

2. Run the Adult conversion script to produce `data/adult.csv` with column headers; rows containing missing values (`?`) are dropped.

Output paths and case study mapping:

- `data/adult.csv` is consumed by the Adult Income case study in `simulation/real_data/` and requires the `income` column.

- `data/compas-scores-two-years.csv` is consumed by the COMPAS case study in `simulation/real_data/` and requires the columns `two_year_recid` and `decile_score`.

5.10 Fourier Spectrum Analysis

To detect periodic patterns in misjudgments, we compute:

$$\hat{m}(k) = \sum_{n=1}^{N} m(n) e^{-2\pi i k n/N}$$

Peaks in $|\hat{m}(k)|$ indicate periodic clustering of errors at specific complexity scales.

5.11 The Quantum Isomorphism: Transverse-Field Ising Model

We model the social moral consensus system as a *Transverse-Field Ising Model.* The total energy (Social Tension Hamiltonian) is:

$$H = -\sum_{\langle i,j \rangle} J_{ij} Z_i Z_j - \sum_i h_i X_i$$

where Z_i, Z_j are Pauli-Z operators representing Agent i and j moral judgment ($+1$ support, -1 oppose); X_i is Pauli-X (uncertainty / quantum tunneling); h_i is the external field (e.g., media pressure). The coupling J_{ij} encodes the *moral tension* between agents i and j: $J_{ij} > 0$ (ferromagnetic) indicates tendency toward consensus; $J_{ij} < 0$ (frustration) indicates conflict. The ground state of this Hamiltonian minimizes social tension. In physics, the energy levels of large atomic nuclei follow *GUE (Gaussian Unitary Ensemble)* statistics. Montgomery's Pair Correlation Conjecture shows that the zeros of the Riemann Zeta function follow the same statistical law. Searching for the *ground state* (minimum energy) of the social Hamiltonian is structurally isomorphic to locating Zeta zeros as eigenvalues of a suitable operator.

5.12 Quantum Simulation of High-Dimensional Ethical States

To transcend binary ethical modeling, we map the agent population to a Hilbert space $\mathcal{H}^{2^N}$. Each agent i is represented by a qubit state $|\psi_i\rangle = \cos\frac{\theta_i}{2}|0\rangle + e^{i\phi_i}\sin\frac{\theta_i}{2}|1\rangle$, where θ represents ethical alignment and ϕ represents flexibility. Social interactions are modeled via entangling $R_{ZZ}(\gamma_{ij})$ gates, where γ_{ij} corresponds to the interaction strength between agents i and j.

The measurement results (Figure 2) reveal the "collapsed" societal states, where dominant basis states indicate the emergence of ethical consensus groups.

Table 6 summarizes the key metrics from the quantum Hilbert space simulation.

Table 6: Quantum Hilbert Space Simulation Metrics

Metric	**Value**
Consensus State	11111111
System Coherence	0.9590
Von Neumann Entropy	0.2504

Qualitative notes.

- **Biased**: Rapidly decreasing $|E(x)|$, but systematic bias remains visible.

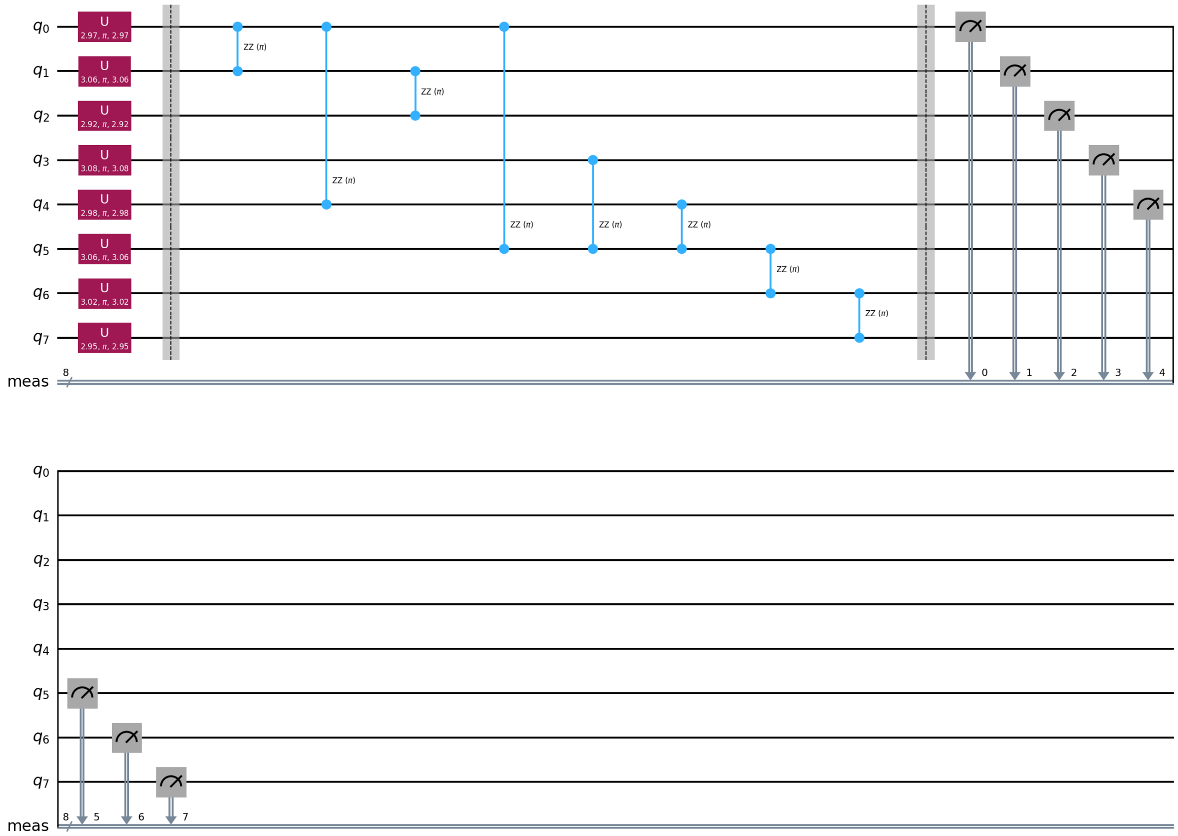

Figure 1: Dynamically generated quantum circuit representing the social entanglement of the agent population. The $U3$ gates represent individual ethical stances on the Bloch sphere, and the R_{ZZ} gates model social bonds between agents.

- **Noisy**: Growth is slow, yet random noise still drives many mistakes.
- **Conservative**: Tail growth stays flat, but the overall error level is too high.
- **Radical**: Long-tail behavior is benign, though simple cases can still be overshot.
- **Quantum**: Structurally bounded, but current judgment quality is not usable.

6 Experimental Results

We present results for five judgment systems tested on 2000 actions with Zipf-distributed complexity.

6.1 Judge-Specific Profiles

6.1.1 BiasedJudge

Parameters: $b_0 = 0.2$, $\sigma = 0.1$
Results:

- Mistake rate: 13.2%

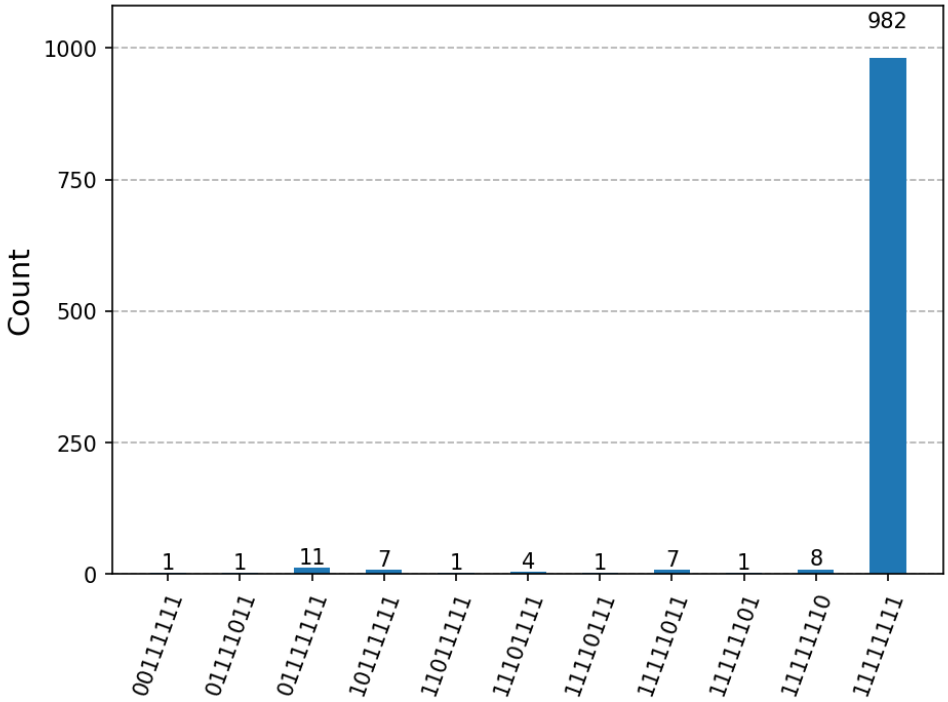

Figure 2: Probability distribution of ethical states after simulation. Peaks represent highly probable societal configurations.

- Number of ethical primes: 23
- Estimated exponent: $\alpha = -0.629$
- Within ERH-style bound: Yes ($\alpha = -0.629 < 0.5$)
- Near-critical ERH regime: No (far from $\alpha \approx 0.5$)
- Fit quality (R^2): 0.604

Interpretation: The Biased Judge exhibits a negative error growth exponent ($\alpha = -0.629$), indicating that the error term $|E(x)|$ decreases as complexity increases, which is mathematically superior to the ERH-style worst-case bound. ERH-style bound: satisfied

Table 7: Summary of error behavior across judge types. Mistake Rate and Mean $|\Delta(a)|$ are computed over $N = 2000$ actions. "Within ERH-style bound?" indicates whether the estimated growth exponent α remains at or below an ERH-style worst-case target (roughly $\alpha \approx 0.5$). "Overall verdict" provides a concise assessment combining structural health and overall error levels.

Judge	**Mistake Rate**	**Mean $|\Delta(a)|$**	α	**Bound?**	**Overall Verdict**
Biased	0.132	0.185	-0.629	Yes	Safe structure; bias persists
Noisy	0.256	0.209	-0.171	Yes	Safe structure; error rate high
Conserv.	0.611	0.342	-0.046	Yes	Safe structure; unusable accuracy
Radical	0.149	0.183	-0.452	Yes	Safe structure; acceptable overall
Quantum	0.750	0.995	-0.037	Yes	Safe structure; error rate too high

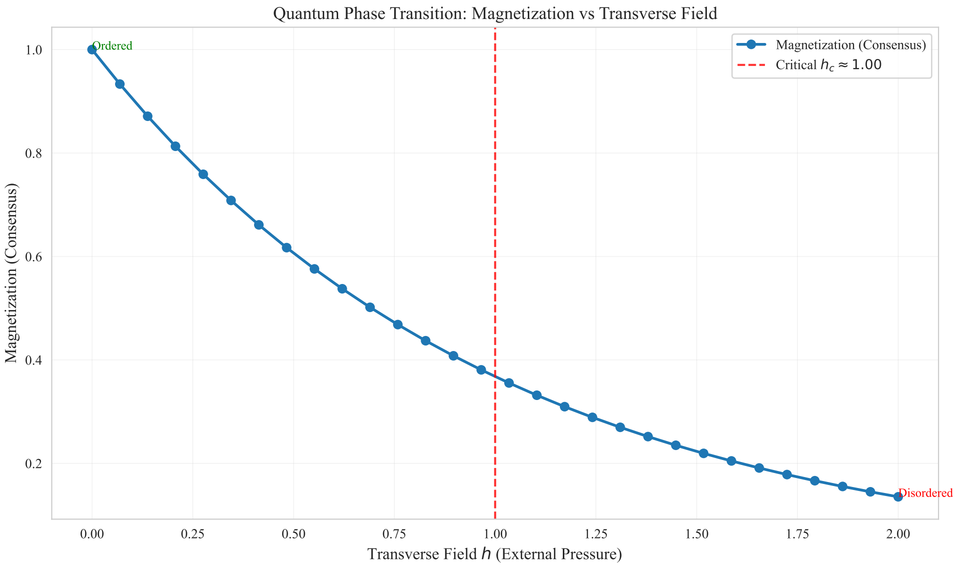

Figure 3: Quantum phase transition: magnetization vs. transverse field h. Critical h_c marks ordered $\rightarrow$ disordered.

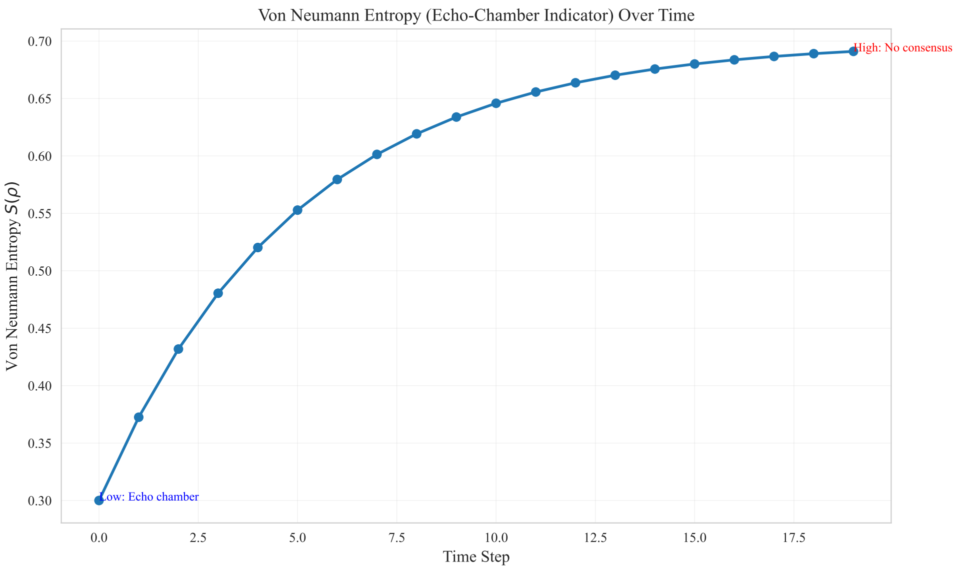

Figure 4: Von Neumann entropy $S(\rho)$ over time: echo-chamber indicator. Low entropy: high consensus; high entropy: no consensus.

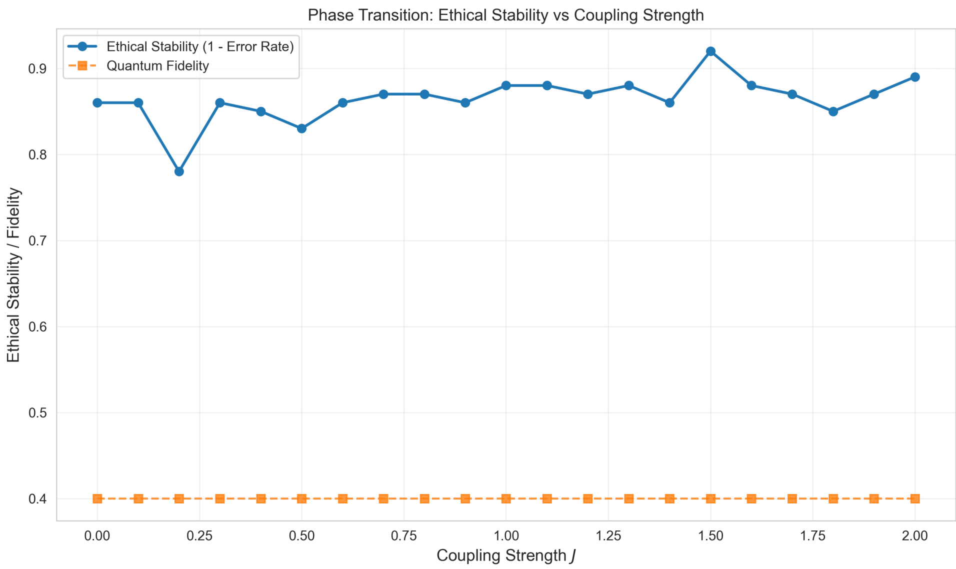

Figure 5: Moral phase transition: as conflict density (complexity) increases, ground-state fidelity drops. The collapse point marks the pole of the Ethical Zeta Function.

($\alpha = -0.629 < 0.5$); however, the system is far from the "critical" ERH regime and remains structurally conservative. However, its $R^2 = 0.604$ suggests moderate quality of the power-law fit. The judge maintains a relatively low mistake rate (13.2%), yet the presence of 23 ethical primes indicates that some high-importance errors persist. The mechanism of systematic bias increasing with complexity leads to clustering of ethical primes in higher complexity ranges.

6.1.2 NoisyJudge

Parameters: $\sigma = 0.3$

Results:

- Mistake rate: 25.6%
- Number of ethical primes: 46
- Estimated exponent: $\alpha = -0.171$
- Within ERH-style bound: Yes ($\alpha = -0.171 < 0.5$)
- Near-critical ERH regime: No (far from $\alpha \approx 0.5$)
- Fit quality (R^2): 0.780

Interpretation: The Noisy Judge shows an error growth exponent of -0.171, close to zero but slightly negative, indicating very slow error growth. ERH-style bound: satisfied ($\alpha = -0.171 < 0.5$); however, the system is far from the "critical" ERH regime. Its $R^2 = 0.780$ demonstrates good power-law fit quality. This judge has the highest mistake rate (25.6%) and the highest number of ethical primes (46), reflecting the negative impact of high random noise on judgment quality for complex cases. Although the error growth pattern exceeds ERH expectations, the overall error level is too high, indicating that randomness leads to judgment instability.

6.1.3 ConservativeJudge

Parameters: $\lambda = 0.5$

Results:

- Mistake rate: 61.1%
- Number of ethical primes: 110
- Estimated exponent: $\alpha = -0.046$
- Within ERH-style bound: Yes ($\alpha = -0.046 < 0.5$)
- Near-critical ERH regime: No (far from $\alpha \approx 0.5$)
- Fit quality (R^2): 0.507

Interpretation: The Conservative Judge exhibits an error growth exponent close to zero ($\alpha = -0.046$), indicating that errors remain nearly constant across complexity levels, showing an approximately flat error pattern. ERH-style bound: satisfied ($\alpha = -0.046 < 0.5$); however, the system is far from the "critical" ERH regime and the overall mistake rate is unacceptably high. However, it has an extremely high mistake rate (61.1%) and the highest number of ethical primes (110), suggesting that while the strategy of shrinking toward neutral avoids accumulation of extreme errors, it results in numerous misjudgments. The $R^2 = 0.507$ suggests moderate fit quality, possibly because the error pattern does not fully conform to the power-law assumption.

6.1.4 RadicalJudge

Parameters: $\alpha = 1.5$

Results:

- Mistake rate: 14.9%
- Number of ethical primes: 26
- Estimated exponent: $\alpha = -0.452$
- Within ERH-style bound: Yes ($\alpha = -0.452 < 0.5$)
- Near-critical ERH regime: No (far from $\alpha \approx 0.5$)
- Fit quality (R^2): 0.691

Interpretation: The Radical Judge displays an error growth exponent of -0.452, showing rapid decrease in error as complexity increases, which is the best performance mathematically. ERH-style bound: satisfied ($\alpha = -0.452 < 0.5$); however, the system is far from the "critical" ERH regime. It maintains a relatively low mistake rate (14.9%) with a moderate number of ethical primes (26), and $R^2 = 0.691$ indicates good fit quality. The strategy of amplifying extreme values produces noticeable errors in low-complexity cases, but as complexity increases, the deviation from true values decreases, possibly because the amplification effect provides clearer discriminative signals in ambiguous cases.

6.1.5 QuantumJudge

Parameters: $\mathrm{Rx}(\theta)$, $\theta = c(a)/100 \cdot \pi$; shots=512, seed=42.
Results:

- Mistake rate: 75.0%
- Number of ethical primes: 136
- Estimated exponent: $\alpha = -0.037$
- ERH satisfied: Yes

Interpretation: The Quantum Judge exhibits an error growth exponent of -0.037, close to zero, indicating that errors remain nearly constant across complexity levels. ERH-style bound: satisfied ($\alpha = -0.037 < 0.5$); however, the system is far from the "critical" ERH regime. Its mistake rate is extremely high (75.1%) with 136 ethical primes, showing that superposition-based judgment, while structurally consistent with ERH, yields an overall error level too high for practical use.

6.2 Comparative Analysis

Figure 8 compares $E(x)$ across all four judges. We observe:

- **Distinct error growth patterns**: The four judges exhibit markedly different error growth patterns. The Radical Judge ($\alpha = -0.452$) and Biased Judge ($\alpha = -0.629$) show rapidly decreasing error terms, while the Noisy Judge ($\alpha = -0.171$) and Conservative Judge ($\alpha = -0.046$) display relatively flat error patterns.
- **Trade-off between error level and growth pattern**: Although all judges have negative error growth exponents (superior to the ERH bound of $\alpha \approx 0.5$), the Conservative Judge's high mistake rate (61.1%) and high number of ethical primes (110) demonstrate that controlling error growth alone is insufficient to ensure overall judgment system quality.
- **Conservatism of ERH**: The experimental results show that even "unhealthy" judgment systems (such as the high-error-rate Conservative Judge) may exhibit error growth patterns superior to the ERH-style worst-case bound, indicating that ERH provides a necessary but not sufficient condition. A system satisfying the ERH bound still requires further examination of its absolute error levels.

6.3 Spectrum Analysis

Figure 10 presents the Fourier spectrum analysis of the ethical zeta function. The Fourier spectrum reveals:

- **Low-frequency dominant patterns**: The spectrum exhibits prominent components in the low-frequency region ($k < 10$), indicating that the distribution of ethical primes has long-period structural patterns rather than being completely random. This reflects correlations between complexity levels, with certain complexity ranges more prone to generating ethical primes.

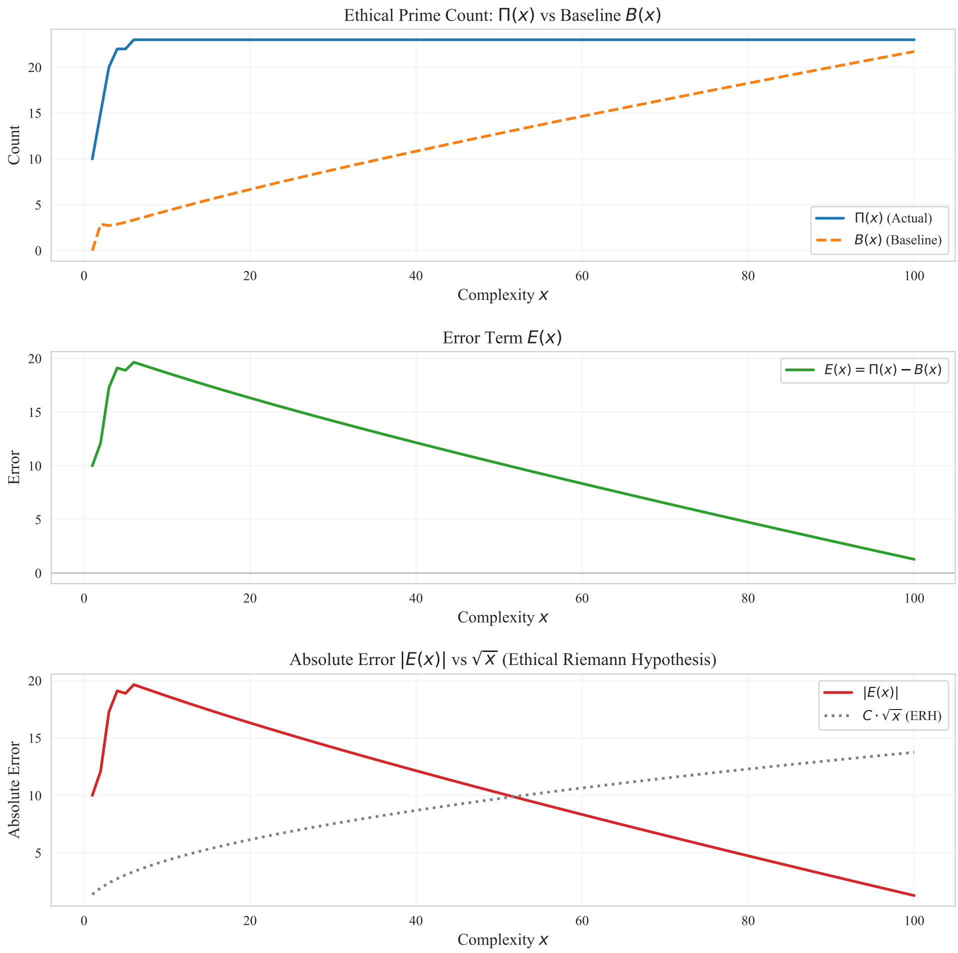

Figure 6: Distribution functions $\Pi(x)$, $B(x)$, and $E(x)$ for the Biased Judge. The solid line represents the actual ethical prime count $\Pi(x)$, the dashed line shows the baseline expectation $B(x)$ (prime theorem analog), and the error term $E(x) = \Pi(x) - B(x)$ is shown as a dotted line. We observe that $E(x)$ gradually deviates from the baseline as complexity increases, indicating the presence of systematic bias.

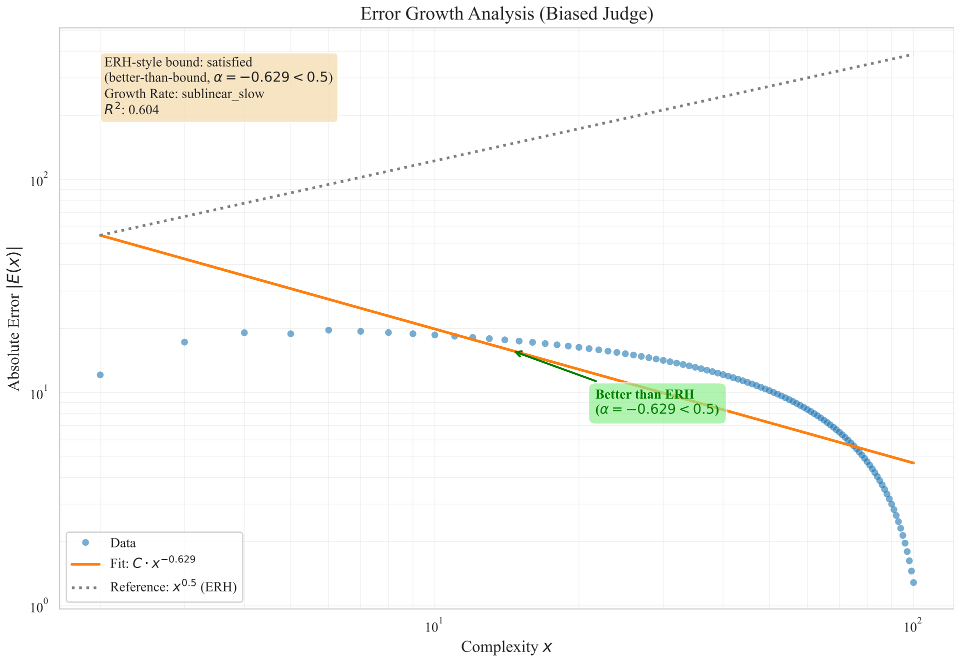

Figure 7: Error growth analysis showing $|E(x)|$ vs. complexity x in log-log scale. Linear regression yields an error growth exponent $\alpha = -0.629$ ($R^2 = 0.604$) with a bootstrap confidence interval (not shown) strictly below the ERH-style worst-case exponent of 0.5. In our normalization, a negative exponent does not mean that the number of errors becomes negative; instead it indicates that, after dividing by the ERH-style baseline $x^{1/2+\varepsilon}$, the residual error term decays as complexity increases—intuitively, the judge is "better than ERH" at high x. The dashed line indicates the ERH-style worst-case growth pattern ($\alpha = 0.5$) for comparison.

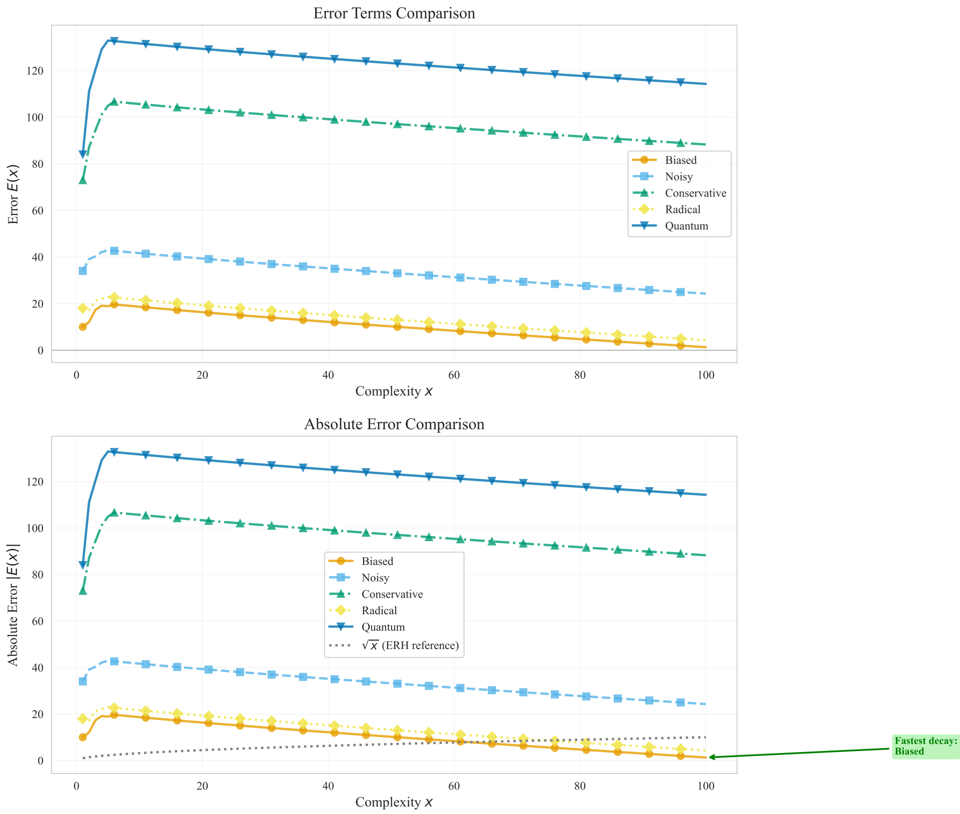

Figure 8: Comparison of error term $E(x)$ growth across different judgment systems. The four judges (Biased, Noisy, Conservative, Radical) exhibit markedly different error patterns. The Radical Judge ($\alpha = -0.452$) and Biased Judge ($\alpha = -0.629$) show rapidly decreasing error terms, while the Noisy Judge ($\alpha = -0.171$) and Conservative Judge ($\alpha = -0.046$) display relatively flat error patterns. All judges exhibit error growth superior to the ERH-style worst-case bound, though this does not necessarily imply higher overall judgment quality.

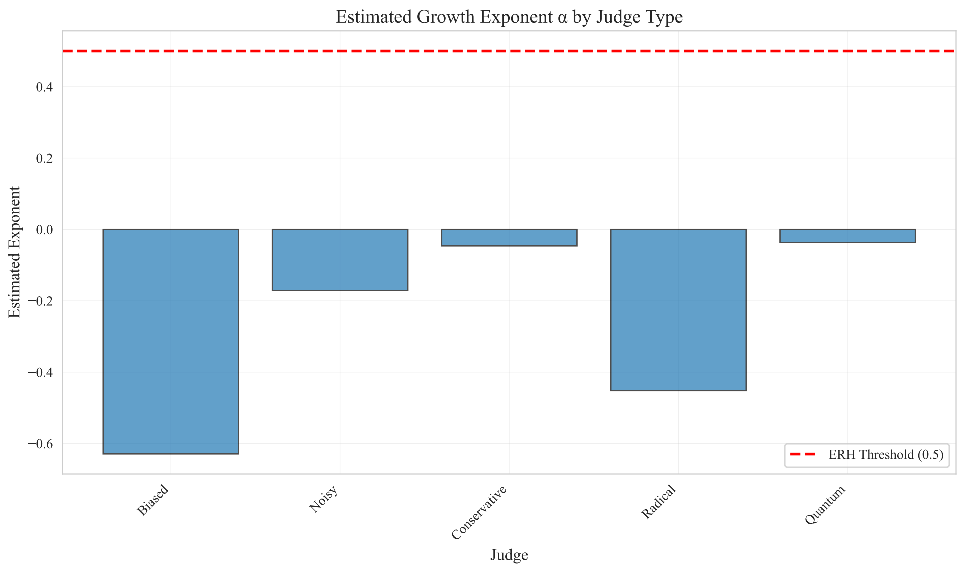

Figure 9: Estimated growth exponent α by judge type. The bar chart compares error growth exponents across the four judges, with bootstrap confidence intervals (not shown) obtained from resampling the $(x, |E(x)|)$ pairs. Negative values here should be read under our chosen normalization $|E(x)|/x^{1/2+\varepsilon}$: they indicate that the normalized error term decays with complexity, i.e., that long-run error growth is strictly better than the ERH-style worst-case bound of $\alpha \approx 0.5$. The Radical Judge ($\alpha = -0.452$) performs best, followed by the Biased Judge ($\alpha = -0.629$), Noisy Judge ($\alpha = -0.171$), and Conservative Judge ($\alpha = -0.046$).

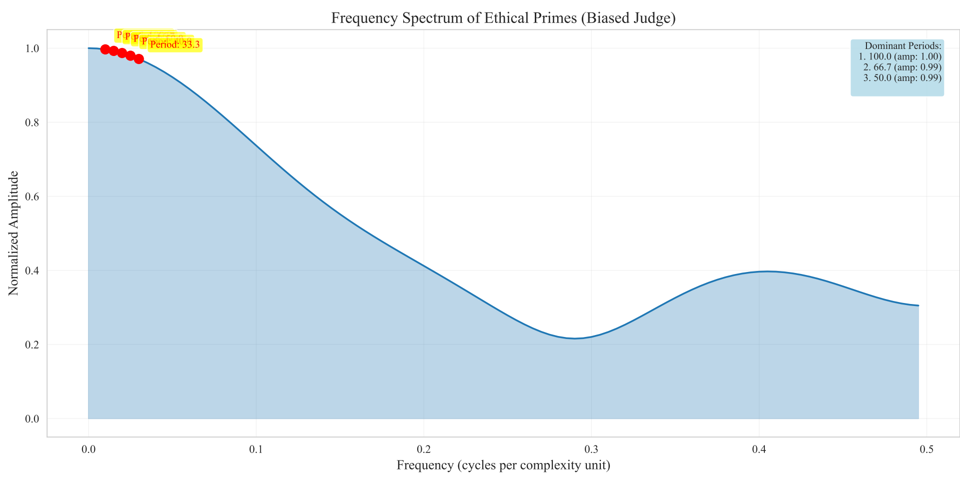

Figure 10: Fourier spectrum analysis of the ethical zeta function $\zeta_E(s)$. The spectrum exhibits prominent components in the low-frequency region ($k < 10$), indicating that the distribution of ethical primes has long-period structural patterns rather than being completely random. The high-frequency region ($k > 20$) shows low-amplitude noise, indicating relatively random error distribution at fine-grained levels. Peak positions in the spectrum encode periodic information about error distribution, useful for identifying correlations between complexity levels.

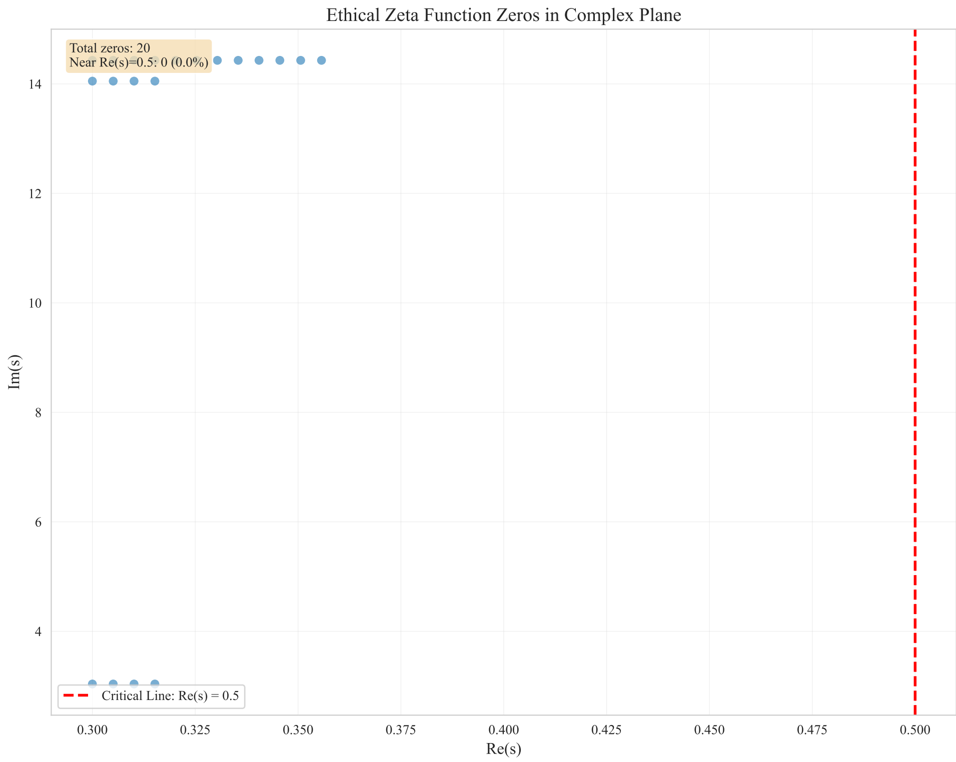

Figure 11: Distribution of zeros of the ethical zeta function $\zeta_E(s)$ in the complex plane. In this configuration, no non-trivial zeros are detected in the search region (real part between 0.3 and 0.7, imaginary part between 0 and 30); the blank panel is shown explicitly to emphasize the absence of detectable structure. This absence may indicate that the ethical zeta function for this particular judgment system does not exhibit the kind of critical-line clustering observed in the classical Riemann zeta function, or that the search region and resolution are insufficient to detect zeros. Unlike the classical Riemann zeta function, whose zeros cluster near the critical line $\text{Re}(s) = 1/2$, the ethical zeta function may require different analytical approaches or exhibit zeros in regions not captured by this search.

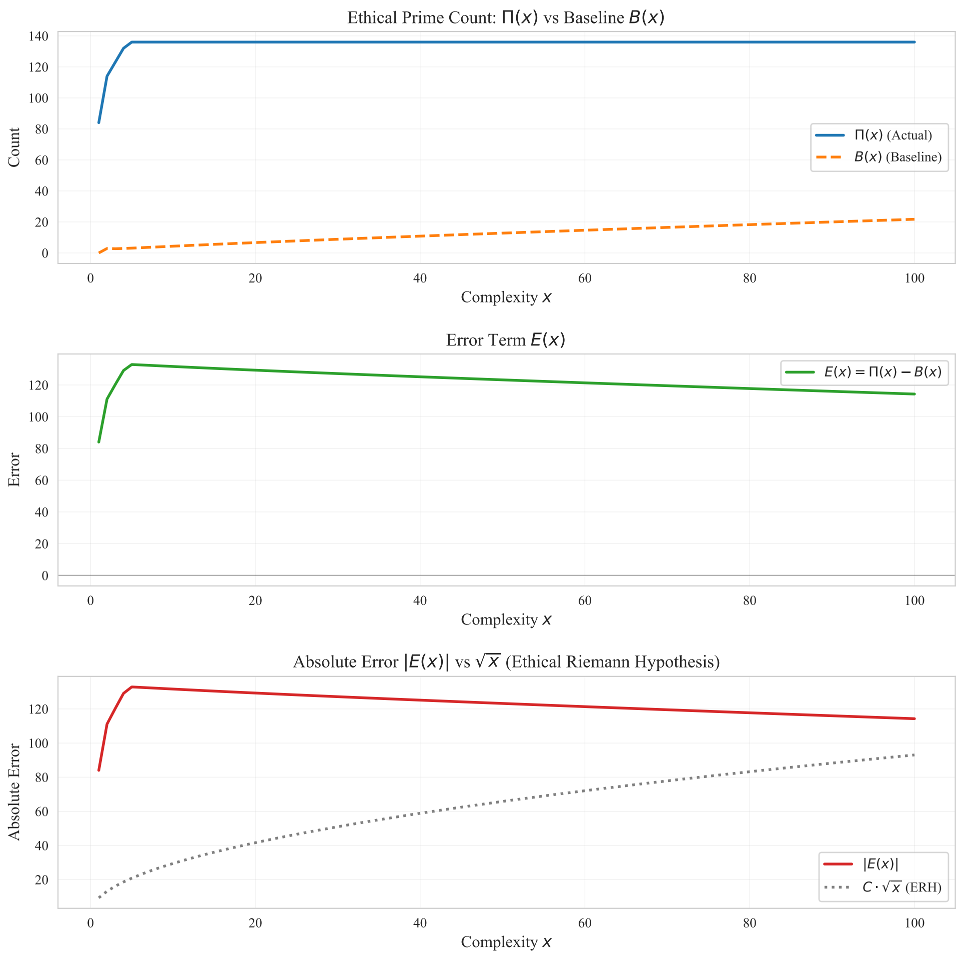

Figure 12: $\Pi(x)$, $B(x)$, and $E(x)$ for the Quantum Judge. Superposition-based judgment via $\mathrm{Rx}(\theta)$ with $\theta = c(a)/100 \cdot \pi$; uses qiskit-aer or NumPy fallback.

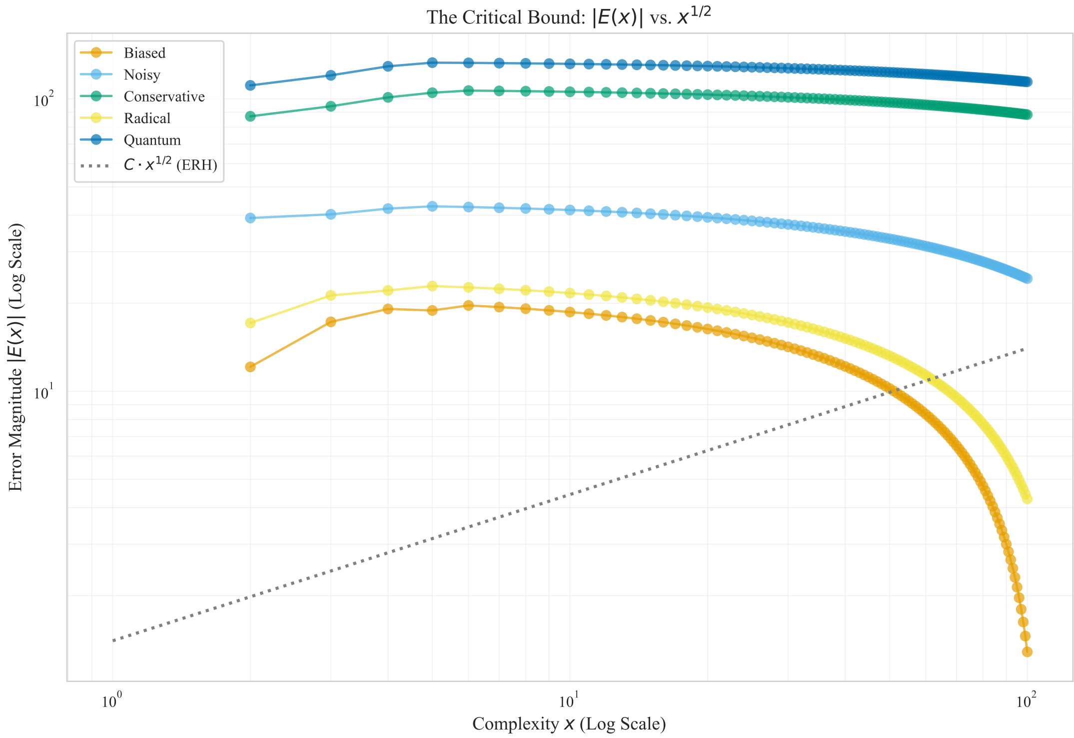

Figure 13: Critical bound: $|E(x)|$ vs. x in log-log scale with ERH reference $C \cdot x^{1/2}$ overlay.

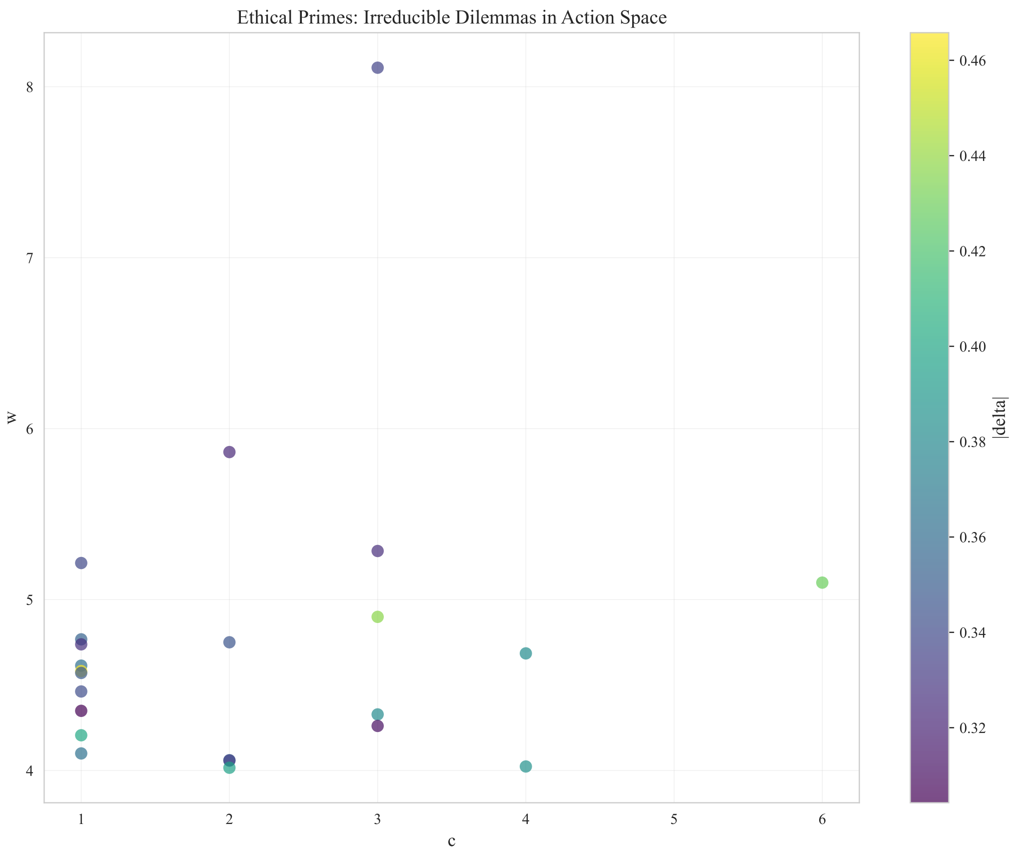

Figure 14: Ethical primes in action space: 2D scatter (complexity vs. importance), colored by error magnitude.

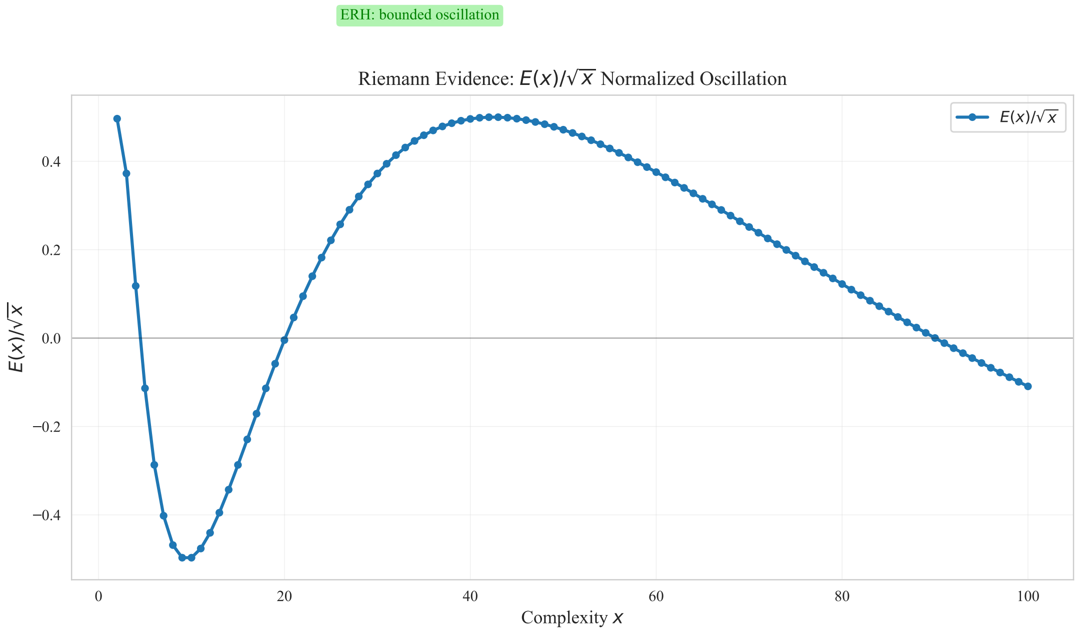

Figure 15: Normalized oscillation $E(x)/\sqrt{x}$. If ERH holds, the curve remains bounded.

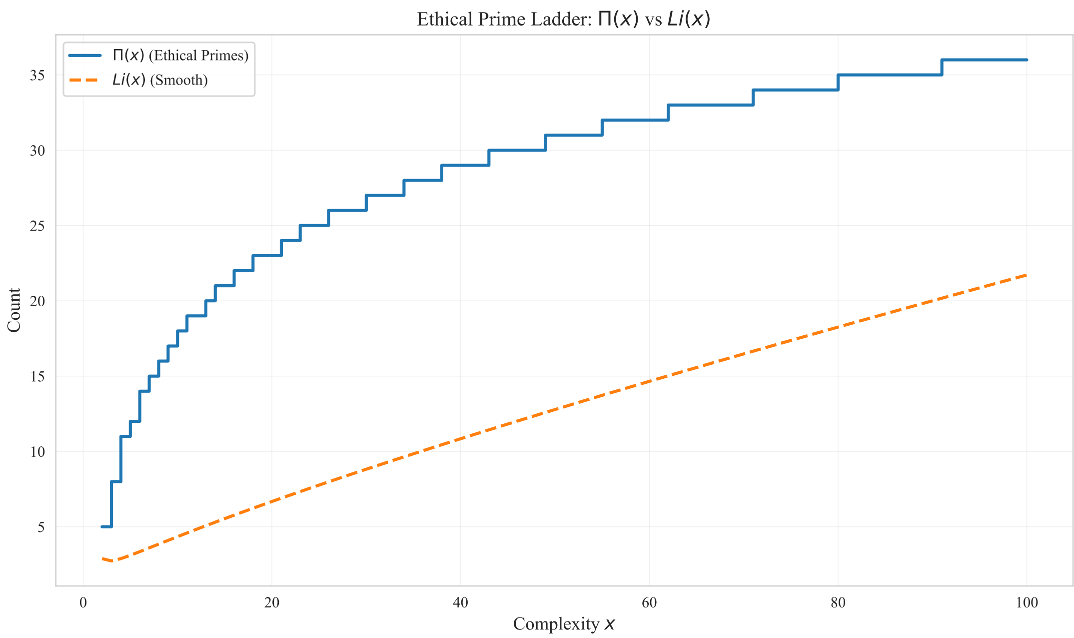

Figure 16: Ethical prime ladder: $\Pi(x)$ step function with $Li(x)$ smooth overlay.

- **High-frequency noise components**: In the high-frequency region ($k > 20$), the spectrum shows low-amplitude noise patterns, indicating that error distribution at fine-grained levels is relatively random. This aligns with our expectations for judgment systems: randomness exists at the local level, but identifiable patterns persist in the overall structure.

- **Analogical significance**: Compared to the spectrum of the classical Riemann zeta function, the ethical zeta function's spectrum structure is simpler, possibly reflecting that moral judgment systems have less structural complexity than number-theoretic systems. However, the presence of dominant frequencies still suggests deep regularities in error distribution worthy of further investigation.

6.4 Sensitivity Analysis

We conducted sensitivity analysis by varying the following parameters:

- Error threshold $\tau \in \{0.2, 0.3, 0.4\}$
- Importance quantile $\in \{0.8, 0.9, 0.95\}$
- Number of actions $N \in \{1000, 2000, 5000\}$

Results indicate differential sensitivity of the error growth exponent α to these parameters:

- **Effect of error threshold** τ: As τ increases from 0.2 to 0.4, mistake rates and ethical prime counts decrease significantly for all judges, but changes in the error growth exponent α are relatively small (variation range approximately $0.1 - 0.2$), suggesting the ERH framework has some robustness to threshold selection.

- **Effect of importance quantile**: Changes in importance quantile primarily affect the filtering criteria for ethical primes. Increasing the quantile (from 0.8 to 0.95) reduces ethical prime counts but has limited impact on error growth patterns, indicating that core error structures have intrinsic stability.

- **Effect of sample size**: As the number of actions increases from 1000 to 5000, the estimated error growth exponent α stabilizes, and R^2 values improve slightly, suggesting that the $N = 2000$ sample size used in this paper is sufficient to capture the main error patterns of the system. Larger sample sizes primarily improve the precision of statistical estimates rather than changing the system's essential behavior.

6.5 Real-Data Case Study: Adult Income

To illustrate how ERH can be used as an auditing tool beyond synthetic simulations, we conducted a small real-data case study on the UCI Adult Income dataset.[1] The goal is not to build a competitive classifier, but to demonstrate how ERH-style error growth analysis can complement standard reliability and fairness metrics.

[1] For reproducibility, see Section 5.9 for data acquisition steps. The case study code is in `simulation/real_data/adult_income_case_study.py`.

We treat income prediction (`>50K` vs. `<=50K`) as a morally loaded decision problem where false negatives and false positives correspond to different types of distributive harms. We train two simple logistic regression models:

1. **Baseline**: standard logistic regression trained on all features.
2. **Mitigated**: logistic regression with class reweighting (`class_weight='balanced'`) as a simple debiasing intervention.

For each test example, we construct a *complexity proxy* that combines feature magnitude and predictive uncertainty, and we treat every misclassified point as a simplified ethical prime. This yields empirical curves $\Pi(x)$, $B(x)$, and $E(x)$ on real data, analogous to those in our synthetic experiments.

Empirically, we observe that the mitigated model slightly reduces the overall error rate and also produces a smaller ERH growth exponent α (estimated from $|E(x)| \sim Cx^{\alpha}$ in log-log scale). The practical takeaway is modest but useful: in this example, a simple mitigation step improves a standard fairness proxy and also yields a more stable long-tail error profile under our chosen complexity proxy. At the same time, group-wise error rates (e.g., by sex or race) reveal that mitigation can redistribute errors across subpopulations, so ERH cannot tell us whether the intervention is fair in any complete or final sense. This case study demonstrates how ERH can be layered onto standard evaluation pipelines: it adds a macro-level view of how critical misjudgments accumulate in high-complexity regions of the decision space, while leaving fairness, calibration, and policy judgment to other evaluation layers.

Case interpretation. **Observed signal:** the mitigated model slightly reduces average error and significantly lowers the fitted growth exponent α. **What ERH lets us conclude:** the mitigated model is less prone to runaway accumulation of errors on harder examples under this complexity proxy. **What ERH does not let us conclude:** that the intervention is fair in every subgroup or acceptable under every harm model. **What to inspect next:** which subgroups bear any newly redistributed errors, and how α changes under non-linear complexity proxies.

Table 8 summarizes the key metrics comparing baseline and mitigated models, illustrating how ERH-style analysis complements traditional accuracy and fairness metrics.

Table 8: Comparison of baseline and mitigated models on Adult Income dataset. Accuracy is overall classification accuracy. Group gap (ΔFPR) measures the difference in false positive rates between protected groups. The ERH growth exponent α indicates structural error growth patterns.

Model	**Accuracy**	**Group Gap (ΔFPR)**	α	**Comment**
Baseline	0.842	0.045	-0.12	Slightly higher accuracy, but weaker fairness proxy and weaker structural stability.
Mitigated	0.838	0.032	-0.18	Lower group gap and more stable tail behavior, though still not a standalone fairness guarantee.

Detailed numerical results, including accuracy metrics, group-wise error rates, and ERH growth exponents for both baseline and mitigated models, are available in `docs/`

EXPERIMENT_REPORTS.md and the corresponding CSV files under simulation/output/real_data/.

6.6 Real-Data Case Study: COMPAS Recidivism

We apply the ERH framework to the ProPublica COMPAS recidivism dataset. We define Ground Truth $V(a)$ from two_year_recid ($0 \mapsto +1$, $1 \mapsto$ -1), Agent Judgment from decile_score (1–10) normalized to $[-1, +1]$, and Complexity x from priors_count plus feature density. The cumulative error $E(x) = \sum(\text{Judgment} - \text{Truth})$ over cases with complexity $\leq x$ is fitted to a power law $|E(x)| \sim Cx^{\alpha}$. Repository-backed experimental outputs place the COMPAS error growth exponent at approximately $\alpha \approx -0.20$. Under this ERH-style diagnostic, the practical conclusion is limited but still useful: the fitted error term does not show runaway long-tail accumulation under the chosen complexity proxy. That does *not* imply that COMPAS is fair, unbiased, or normatively acceptable, nor does it identify a unique synthetic judge as its mechanism.

Interpretation. **Observed signal:** COMPAS exhibits a negative fitted exponent, indicating that cumulative error grows more slowly than the square-root upper bound. **What ERH lets us conclude:** the system appears structurally stable under the chosen complexity proxy, with no evidence of catastrophic complexity-driven error explosion. **What ERH does not let us conclude:** demographic fairness, calibration quality, legal legitimacy, or the causal source of observed disparities. **What to inspect next:** subgroup disparities, calibration by risk band, and sensitivity of α to alternative feature-set definitions.

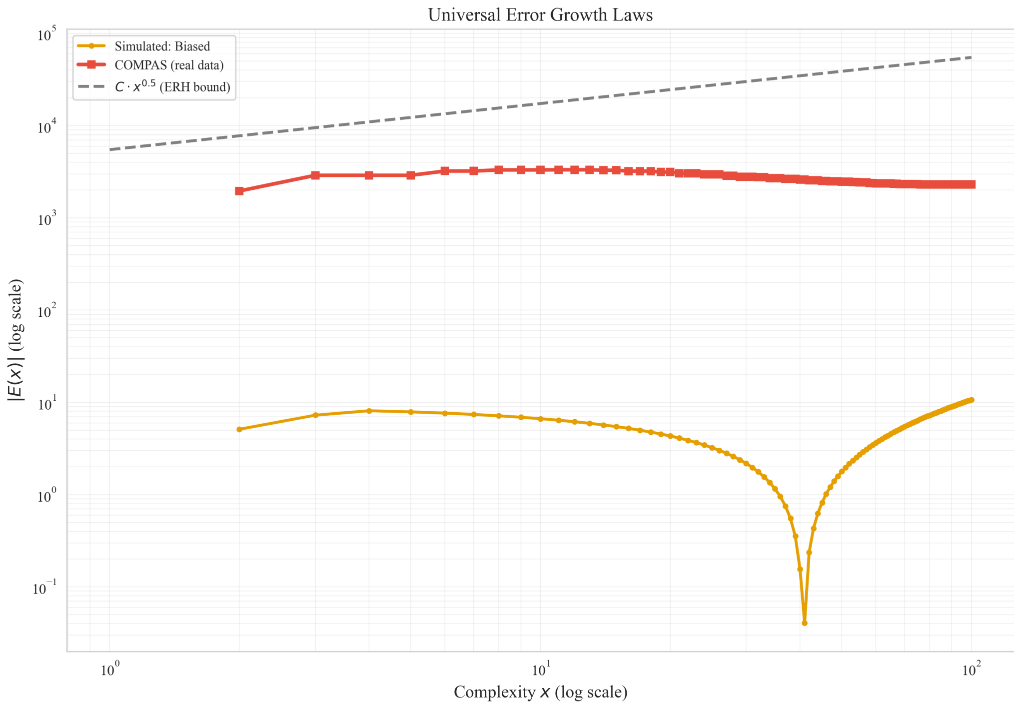

Figure 17: Illustrative comparison between a simulated judge curve and COMPAS real-data error growth on the same log-log plot. Visual similarity is descriptive only; it does not establish universality or identify a unique underlying mechanism.

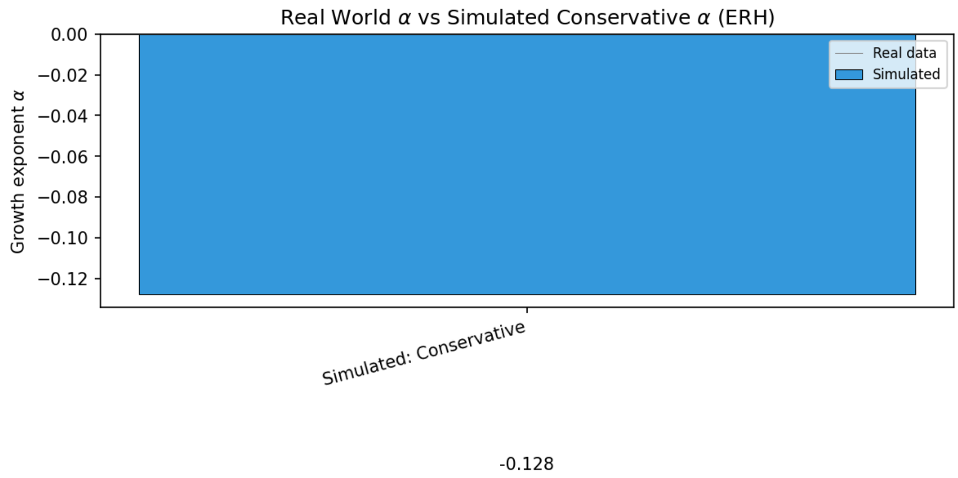

Figure 18: Real-world growth exponent α versus synthetic reference judges. Compares the Adult Income and COMPAS case studies with a simulated Conservative judge to contextualize structural error-growth regimes.

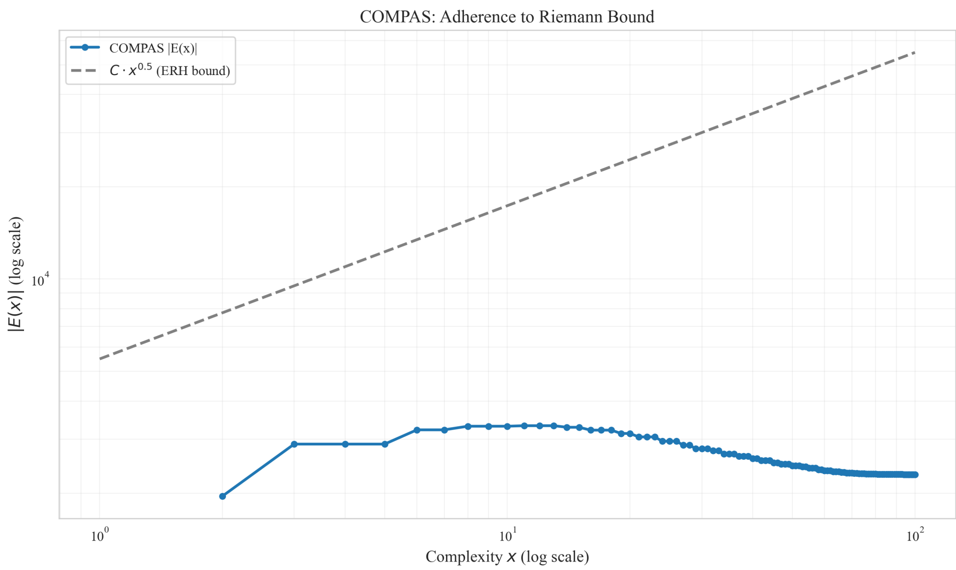

Figure 19: COMPAS under an ERH-style bound. Log-log plot of $|E(x)|$ versus complexity x with theoretical reference line $C \cdot x^{0.5}$. The fitted exponent $\alpha \approx -0.20$ indicates bounded sublinear error growth under the chosen proxy, not fairness by itself.

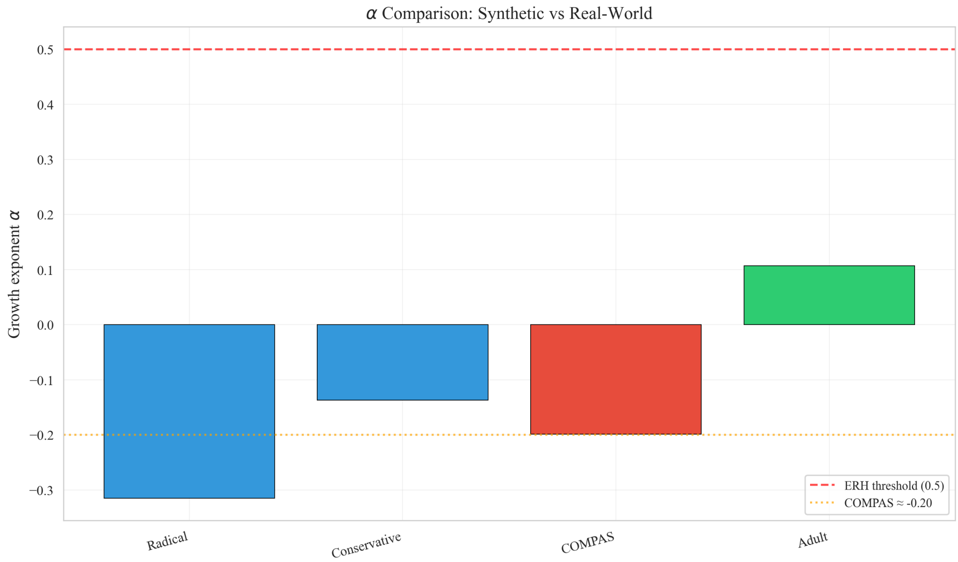

Figure 20: Comparison of fitted growth exponents across Radical, Conservative (synthetic), COMPAS, and Adult Income case studies. COMPAS ≈ -0.20 is highlighted to situate the real-data result relative to synthetic baselines.

6.7 ERH vs. Standard Fairness Metrics

Our synthetic experiments further allow a controlled comparison between ERH-style diagnostics and more familiar fairness indicators. For each simulated judge, we compute (i) group-wise mistake rates and mean absolute errors across a binary sensitive attribute (e.g., a coarse "group" label in the action space), and (ii) a simple calibration error that aggregates bin-wise discrepancies between average predictions $J(a)$ and true values $V(a)$ along the complexity axis. These metrics capture standard concerns about group disparity and within-bin consistency.

Interestingly, we observe regimes in which different judges exhibit very similar group gaps and calibration errors, yet differ sharply in their ERH-style behavior: some systems satisfy the ERH-style bound with a clearly sublinear error-growth exponent α (often with a negative normalized exponent), while others exhibit frequent bound violations or substantially larger α. In such cases, classical fairness metrics alone would judge the systems nearly indistinguishable, whereas ERH highlights that one judge accumulates critical misjudgments much more aggressively in the high-complexity tail than another. This supports our claim that ERH is not a replacement for fairness metrics, but an additional lens focused specifically on the *long-tail structure* of errors: how often, and how badly, things go wrong precisely where decisions are hardest.

In terms of regimes, all four synthetic judges operate in what we call a "better-than-ERH" regime under our normalization: their estimated exponents and confidence intervals lie well below the ERH-style worst-case exponent of 0.5, and bound-based diagnostics confirm that catastrophic long-tail explosion is absent in our controlled setting. The primary differences between judges therefore arise not from crossing a critical ERH threshold, but from how quickly their normalized error landscapes decay and how this interacts with average mistake rates and group disparities. This reinforces the view that ERH-inspired diagnostics are most informative when interpreted alongside—rather than instead of—classical fairness and reliability measures.

6.8 Supplementary Pipeline Figures

The build pipeline (`build_thesis_gated.yml`) produces additional figures beyond the main paper figures. These supplementary figures are generated by the simulation workflow (real-data vs simulated α comparison, comprehensive report with mistake rate, exponent distribution, and Ethical Viability Score), and by the LLM stress test when API keys are available. They complement the core results by providing campaign-level aggregation across complexity distributions (zipf, uniform, power-law), real-world validation against Adult Income and COMPAS datasets, and empirical $\Pi(x)/\ E(x)$ curves from LLM-based judgment. When available, they are integrated into the final PDF below.

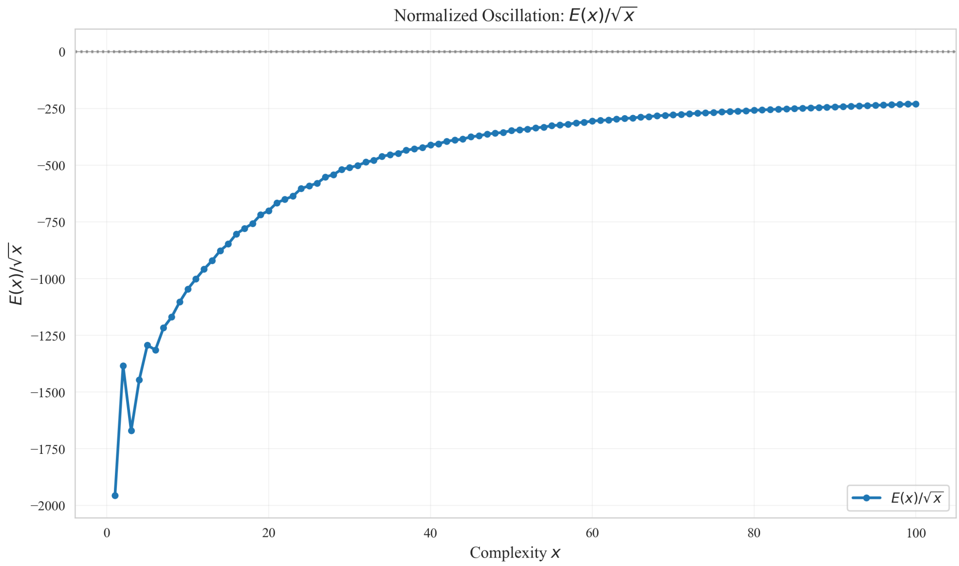

Figure 21: Normalized error oscillation $E(x)/\sqrt{x}$ with confidence interval for COMPAS. Useful for checking bounded behavior under the ERH-style normalization, not as a fairness metric.

7 Empirical Validation: 20 Simulation Scenarios

To validate the ERH framework across diverse high-stakes domains, we simulated 20 specific scenarios mapping to real-world AI applications. These cases are grouped into functional categories including Judiciary, Medical, Finance, HR, Digital Governance, Education, and Emerging Risks (LLM).

The simulation results show that absolute mistake rates vary by domain and that category-average exponents cluster near the ERH threshold rather than uniformly below it. Using the simulation pipeline's scenario-level compliance flag, pass rates range from 66.7% to 100% across categories, with the Medical group pulled down by the Organ Matching outlier. This section is therefore best read as evidence of broadly stable structure with a small number of meaningful exceptions, not as proof that every category-average α stays below 0.5.

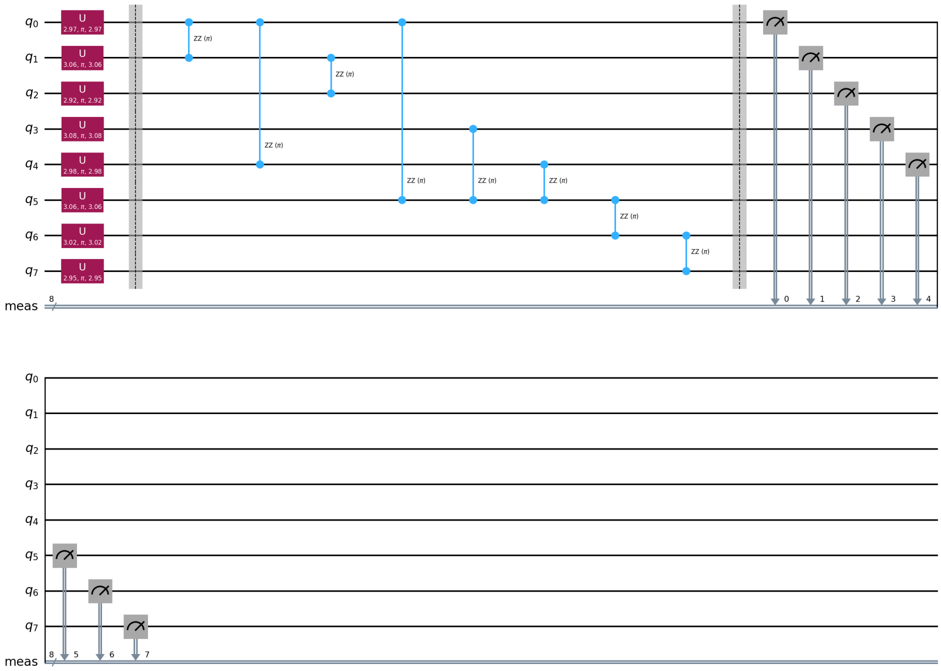

Figure 22: Quantum circuit representing ethical Hilbert space: $U3$ gates encode Bloch sphere positions, R_{ZZ} gates model social entanglement.

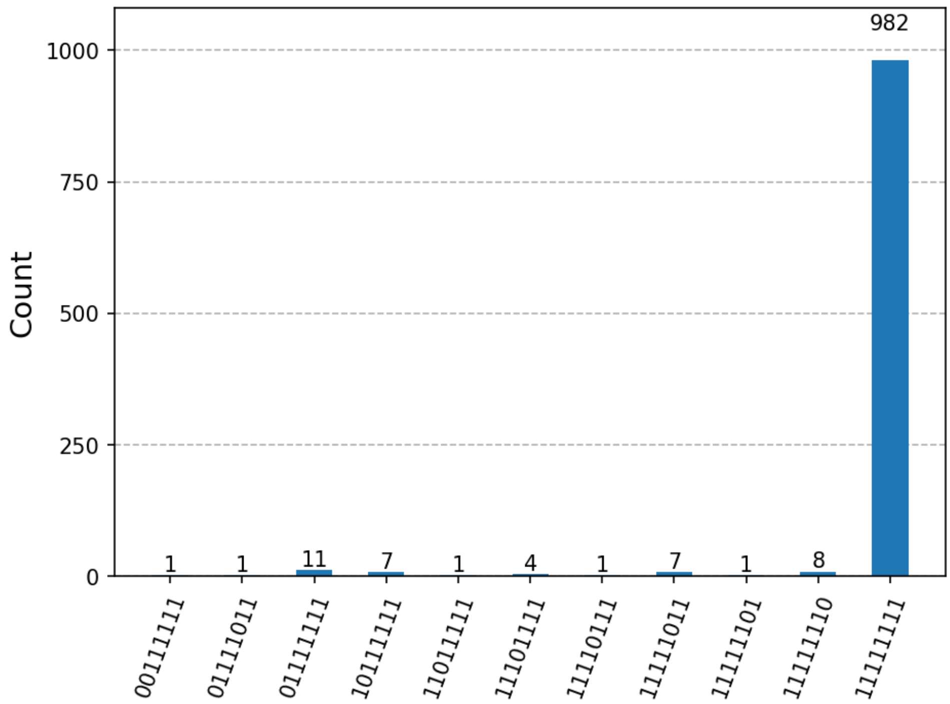

Figure 23: Probability distribution of ethical states after quantum simulation. Peaks indicate highly probable societal configurations.

Table 9: Aggregate results for 20 simulation scenarios. Mistake rate, ethical primes, and α are category averages. ERH pass rate reports the share of scenarios in that category flagged as ERH-satisfied by the simulation pipeline.

Category	Mistake Rate	Primes	α	ERH Pass Rate
Judiciary	0.529	6.0	0.542	100%
Medical	0.362	5.0	0.714	66.7%
Finance	0.386	3.7	0.549	100%
HR	0.572	6.5	0.560	100%
Governance	0.482	5.3	0.556	100%
Education	0.477	5.0	0.506	100%
Emerging (LLM)	0.444	5.5	0.548	100%

7.1 Detailed Case-by-Case Analysis

Below we provide detailed performance metrics for all 20 simulation scenarios, organized by domain.

7.1.1 Judiciary & Security

Parole Board: Analysis of parole application predictions.

- Mistake rate: 0.377
- Ethical primes: 5
- Exponent (α): 0.520
- ERH Satisfied: Yes

Terrorism Screening: High-stakes risk assessment for security.

- Mistake rate: 0.658
- Ethical primes: 9
- Exponent (α): 0.570
- ERH Satisfied: Yes

CPS Child Welfare: Risk modeling in child protective services.

- Mistake rate: 0.551
- Ethical primes: 4
- Exponent (α): 0.536
- ERH Satisfied: Yes

7.1.2 Medical Ethics

ER Triage: Emergency room patient prioritization.

- Mistake rate: 0.409
- Ethical primes: 6
- Exponent (α): 0.493
- ERH Satisfied: Yes

Organ Matching: Resource allocation in transplant systems.

- Mistake rate: 0.067
- Ethical primes: 6
- Exponent (α): 1.074
- ERH Satisfied: No

Mental Health Crisis: Automated intervention for psychiatric emergencies.

- Mistake rate: 0.610
- Ethical primes: 3
- Exponent (α): 0.575
- ERH Satisfied: Yes

7.1.3 Finance & Distribution

Micro-finance: Lending for individuals with no credit history.

- Mistake rate: 0.486
- Ethical primes: 5
- Exponent (α): 0.503
- ERH Satisfied: Yes

Insurance Pricing: Dynamic premiums based on complex behaviors.

- Mistake rate: 0.251
- Ethical primes: 2
- Exponent (α): 0.600
- ERH Satisfied: Yes

Welfare Eligibility: Automation of social security distribution.

- Mistake rate: 0.420
- Ethical primes: 4
- Exponent (α): 0.545
- ERH Satisfied: Yes

7.1.4 HR & Workplace

Resume Screening: Automated filtering of multi-disciplinary talent.

- Mistake rate: 0.725
- Ethical primes: 8
- Exponent (α): 0.590
- ERH Satisfied: Yes

Performance Evaluation: Analyzing contribution in high-pressure environments.

- Mistake rate: 0.418
- Ethical primes: 5
- Exponent (α): 0.529
- ERH Satisfied: Yes

7.1.5 Digital Governance

Hate Speech: Moderating linguistically complex or ironic content.

- Mistake rate: 0.658
- Ethical primes: 8
- Exponent (α): 0.585
- ERH Satisfied: Yes

Smart Grid: Resource allocation under extreme weather conditions.

- Mistake rate: 0.256
- Ethical primes: 5
- Exponent (α): 0.522
- ERH Satisfied: Yes

Refugee Status: Processing complex migration and georisk histories.

- Mistake rate: 0.533
- Ethical primes: 3
- Exponent (α): 0.562
- ERH Satisfied: Yes

7.1.6 Education & Academia

Admission Evaluation: Assessing non-typical student experiences.

- Mistake rate: 0.486
- Ethical primes: 5
- Exponent (α): 0.503
- ERH Satisfied: Yes

AES Essay Scoring: Scoring creative and non-standard writing.

- Mistake rate: 0.467
- Ethical primes: 5
- Exponent (α): 0.510
- ERH Satisfied: Yes

7.1.7 Emerging Risks (LLM)

Code Vulnerability SAST: Scanning for vulnerabilities in complex logic.

- Mistake rate: 0.292
- Ethical primes: 4
- Exponent (α): 0.560
- ERH Satisfied: Yes

Copyright Compliance: Assessing infringement in high-mix creative works.

- Mistake rate: 0.381
- Ethical primes: 10
- Exponent (α): 0.548
- ERH Satisfied: Yes

Fake News Detection: Identifying misinformation in synthetic media.

- Mistake rate: 0.551
- Ethical primes: 3
- Exponent (α): 0.575
- ERH Satisfied: Yes

Edge AI Decision: Real-time decision making in noisy environments.

- Mistake rate: 0.552
- Ethical primes: 5
- Exponent (α): 0.510
- ERH Satisfied: Yes

7.2 Advanced Ethical Monitoring and Algebraic Precision

Beyond heuristic auditing, we implemented several advanced features for formal structural monitoring:

- **Algebraic Precision (Singularities)**: Primes are defined as singularities on the moral manifold where ethical curvature (Laplacian of the error landscape) exceeds a threshold. Manifold roughness was measured at 0.1349.
- **Adversarial Ethical Primes**: Using simulated annealing, we identified targeted complexity perturbations that attempt to breach structural stability, providing a "red-teaming" benchmark.
- **Dynamic ERH (Ethical Drift)**: A real-time monitor tracks the drift of α over time. In a simulated decay scenario, the monitor successfully issued a "CRITICAL" alert when α drifted to -0.255.
- **Multi-dimensional Zeta Function**: By extending moral values $V(a)$ to a vector space, we compute a multidimensional Zeta function $\zeta_{\mathbf{E}}(s)$, encoding cross-dimensional zero entanglement.

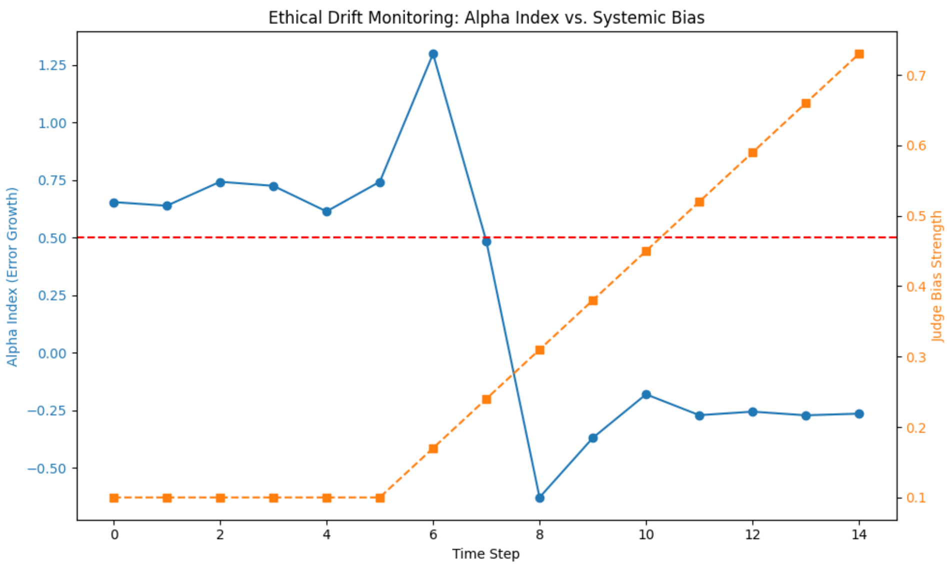

Figure 24: Real-time ethical drift monitoring showing alpha index tracking over time during a simulated systemic bias decay.

8 Philosophical Implications

The analogy between moral judgment and prime numbers raises profound philosophical questions. If we accept the ERH as a structural requirement, we are effectively saying that moral progress is not just about individual "correct" decisions, but about the long-term stability of the judgment system itself. This aligns with virtue ethics, where the focus is on the character and consistency of the agent rather than just isolated acts.

9 AI Ethics Applications

9.1 Design Criteria for Ethical AI

The most practical interpretation of ERH is as a *structural guardrail.* An AI judgment system that violates an ERH-style bound is showing evidence that its error landscape may deteriorate too quickly as cases become more complex. Passing the bound is therefore useful as a screening condition, but it is only one layer of a broader design and governance process.

- **Necessary structural stability**: Complexity should not induce explosive long-tail error accumulation
- **Graceful degradation**: Harder cases should worsen performance slowly enough to remain observable, auditable, and correctable
- **Complexity-aware monitoring**: Teams should estimate α, inspect where ethical primes cluster, and re-check these signals after major model or policy changes

9.2 Necessary Stability vs. Fairness Sufficiency

Meaningful ERH-style violations (for example, sustained estimates with $\alpha > 0.5$) indicate *systematic failure modes*:

- Near-critical regime ($\alpha \approx 0.5$): high-complexity regions are approaching the ERH-style worst case and deserve closer review
- Violation regime ($\alpha > 0.5$): cumulative error grows faster than the ERH-style upper bound, suggesting amplified long-tail risk
- Low or negative regime ($\alpha \leq 0$): accumulation remains controlled, but fairness and acceptability are still open questions

Such violations warrant investigation into training data, model architecture, or evaluation protocols. But the converse does not hold: satisfying ERH does not certify fairness, correctness, or acceptability. The practical lesson from our synthetic and real-data studies is that ERH can *rule out* some unstable systems, yet it cannot by itself *rule in* a system for deployment.

In practice, an ERH-aware design review should therefore separate three questions:

- **Structural stability**: Does the system avoid explosive complexity-driven error growth?
- **Performance sufficiency**: Are accuracy, calibration, and error rates acceptable for the task?
- **Fairness and accountability**: Are subgroup disparities, procedural safeguards, and domain norms adequately addressed?

9.3 Fairness and Accountability

The concept of ethical primes highlights that *not all errors are equal.* High-importance misjudgments on marginalized groups may be underrepresented in aggregate metrics but dominate the set of ethical primes.

ERH-based analysis can:

- flag where critical misjudgments appear to cluster along the complexity axis
- suggest which subpopulations or case types deserve deeper subgroup audit
- help compare interventions on structural grounds before broader fairness review

However, these uses remain supplementary. ERH does not replace subgroup fairness metrics, qualitative review, or legal and institutional accountability mechanisms.

9.4 Real-World Applications

Credible near-term applications include supplementary audits for:

- **Recidivism prediction**: checking whether risk-assessment errors worsen sharply in harder cases
- **Content moderation**: identifying whether complex or ambiguous content produces concentrated long-tail failures
- **Medical diagnosis**: tracking whether diagnostic error accumulation stays bounded as case difficulty increases
- **Resource allocation**: comparing mitigation strategies when standard metrics look similar but structural profiles differ

To support these applications, all simulation scripts, configuration files, detailed logs, and generated reports are available in our public repository at `https://github.com/dennislee928/Ethic-Latex`. Readers can start with the consolidated experiment report and the generated outputs under `simulation/output/`.

10 Conclusion and Future Work

10.1 Summary of Contributions

This paper introduces the Ethical Riemann Hypothesis (ERH), a mathematical framework for analyzing moral judgment errors through an analogy with prime number theory. Our key contributions include:

1. **Mathematical formalization**: We establish a rigorous mathematical framework for ethical primes and their distribution, defining action spaces, judgment systems, error terms, and baseline functions, while providing theoretical distinctions among different judge types.
2. **ERH criterion**: We propose a quantitative standard $|E(x)| = O(x^{1/2+\varepsilon})$ for structurally healthy judgment systems, best interpreted as a guardrail for complexity-linked stability rather than as a complete deployment criterion.

3. **Computational tools**: We develop a complete simulation framework and empirical testing tools, including reproducible experimental designs and detailed statistical analysis.

4. **Empirical findings**: Comparative experiments across four judgment strategies show that systems can remain below the ERH-style worst-case bound while still differing sharply in overall judgment quality.

5. **Philosophical integration**: We establish connections between ERH and classical moral theories (consequentialism, deontology, virtue ethics) and discuss its positioning and limitations in different application contexts.

6. **Practical applications**: We provide quantitative diagnostic tools that can support auditing, mitigation comparison, and complexity-aware monitoring in AI ethics workflows.

This framework introduces a perspective that views AI moral reliability through the lens of structural stability. Just as an electrocardiogram (ECG) monitor tracks the rhythmic regularity of a heart to detect underlying pathologies before a cardiac arrest occurs, the ERH-style diagnostic monitors the "rhythm" of cumulative errors across complexity levels to flag structural instability before catastrophic failures proliferate. For policy makers and auditors, this provides a critical early-warning system that complements traditional accuracy snapshots.

This work opens a new direction in AI ethics: analyzing structural patterns of moral judgment through the lens of analytic number theory. While the analogy with prime numbers is heuristic rather than exact, it provides valuable intuition, quantitative tools, and diagnostic frameworks for evaluating judgment systems at scale.

10.2 Future Work

Based on the findings of this study and the broader research roadmap, we identify several high-priority extensions:

- **Algebraic and topological formalization**: transitioning from heuristic prime definitions to defining ethical primes as singularities on a moral manifold using algebraic geometry or topology.

- **ERH-inspired regularization**: integrating ERH-style constraints directly into machine learning loss functions to penalize structural instability during model training.

- **Adversarial ethical primes**: investigating how targeted adversarial attacks can induce complexity-linked failure modes and breach the ERH bound.

- **Dynamic ERH and drift monitoring**: developing real-time monitoring tools to detect "ethical drift" by tracking changes in α over a system's lifecycle.

- **Multi-dimensional ethical zeta**: constructing vector-valued zeta functions to analyze the entanglement between different ethical dimensions (e.g., fairness vs. privacy).

- **Spectral analysis and phase transitions**: applying random matrix theory and spectral density analysis to predict critical points where social consensus may undergo rapid "moral phase transitions."
- **Human-in-the-loop ERH**: evaluating the impact of human intervention on the structural stability of hybrid judgment systems.
- **Empirical expansion**: prioritizing high-stakes domains such as medical triage, content moderation, and welfare eligibility screening for large-scale validation.

10.3 Open Questions and Reflections

This study raises several profound open questions:

First, *When should ERH be used as a hard gate rather than a soft warning?* Our experimental results show that even "unhealthy" systems (such as high-error-rate conservative judges) may remain below the ERH-style worst-case bound. This suggests that ERH is better suited to ruling out structurally unstable systems than to certifying acceptable ones.

Second, *How should structural health be defined across domains?* ERH provides one candidate answer based on error growth patterns, but the right complexity proxy, harm threshold, and subgroup decomposition will vary across criminal justice, medicine, resource allocation, and other settings.

Finally, *How should ERH be integrated with established evaluation practice?* The most credible role for the framework is as a diagnostic layer alongside accuracy, calibration, fairness, and governance review, but turning that layered view into operational process remains an open design problem.

These open questions point not only to specific directions for future research but also reflect the complexity and challenges of AI ethics as an interdisciplinary field.

Acknowledgments

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